Deep Materials Engineering Analysis, Standards Verification & Industry Report

Basis: uploaded 'mechanical_materials_properties_tests_master_list.xlsx' (132 grades) + master prompt. Prepared 21 Oct 2026.

1. Scope and objectives

Scope

Audit and enrich the supplied 132-grade master list: qualified crystal/grain structure, property database, formula register with independent checks, testing-method register, standards audit, error log, industry considerations and application-based selection. Built as a clearly scoped, source-labelled workbook rather than a claim of full completeness.

2. Key figures (live formulas)

Material grades reviewed (supplied list)	132
Gap-list grades suggested (not counted)	16
Property records (grade/condition rows)	80
Numeric property values recorded	508
Grain / crystal-structure family records	33
Formula blocks / independent checks executed	19
Testing-method rows	44
Standards reviewed (14 supplied + supporting)	27
Errors / corrections logged	22
Industry-consideration entries (advisory)	35

3. Verification status and coverage by domain

Domain	Items reviewed	Verified	Conditionally verified	Incorrect	Unverified	Strict Cv	Conditional Cv	Reconciles?
Master-list designations	132	0	103	3	26	0.0%	78.0%	OK
Property records (Sheet 3)	80	0	54	0	26	0.0%	67.5%	OK
Grain / crystal-structure records	33	0	33	0	0	0.0%	100.0%	OK
Formula checks (Sheet 5)	19	19	0	0	0	100.0%	100.0%	OK
Testing-method references	44	0	35	0	9	0.0%	79.5%	OK
Standards (supplied + supporting)	27	0	25	0	2	0.0%	92.6%	OK
TOTAL	**335**	**19**	**250**	**3**	**63**	**5.7%**	**80.3%**	**OK**

Counting rule

Cv(strict) = N_verified / N_items_reviewed × 100% — 'Verified' means checked against an authoritative document or independently recalculated (all current Verified items are arithmetic/unit checks). Cv(conditional) = (Verified + Conditionally verified) / N_items. Gap-list grades (not in upload) and advisory industry rows are excluded because they are suggestions, not reviewed items. Items are not weighted.

Accuracy vs coverage

Verification coverage is NOT numerical accuracy. Numerical accuracy was not measured (no reference dataset or measurement comparison); property values are recalled handbook/standard values with stated confidence.

Why 90–95% was not reached	No network/authoritative-document access in this session, so standards and datasheet values could not be checked against BIS/ASTM/manufacturer documents. Conditional coverage shows how much is consistent with recalled knowledge; the gap to strict coverage is the remaining verification work (Sheet 11, unresolved items).

4. Major corrections (details in Sheet 8)

• EN8/EN19/EN24 are withdrawn BS 970 designations — replaced by EN 10083 grades (similar, not identical).

• '/' pairs (1045/C45, 4140/42CrMo4, 52100/100Cr6) are similar, not equivalent.

• 'SS 2xx/3xx' is not a standard designation — use ASTM A240 type/UNS/EN 10088/IS 6911.

• IS 2062 E250/E350 are incomplete without quality designation; 17-4 PH needs ageing condition; 2205 has UNS S32205 and S31803.

• ASTM B117 is an apparatus practice (no pass/fail); ASTM E3 is a Guide; ASTM E164 and IS 1865 status unverified.

• Under-specified items (CoCr, graphite, CBN/PCD, composites, ceramics) cannot carry grade-specific properties.

5. Important limitations

• Values are recalled typical/specified figures; none re-fetched; no publication dates recorded.

• Standard editions are recalled and may be outdated; paywalled texts were not examined.

• Fatigue, impact and fracture-toughness values are largely not recorded (grade/condition/test-specific).

• Typical values are not guaranteed minimum design properties; use product-standard minima/design allowables.

• Polymers, ceramics, composites are datasheet/grade dependent — ranges are indicative (Low confidence).

Sheet 2 — Master Material Database (132 supplied grades + reviewer gap list)

One row per supplied designation. Crystal structure/phase text is qualified by condition — no single structure is assigned to a whole alloy family. Statuses: see Executive Summary definitions.

No.	Designation (as supplied)	Corrected / preferred designation	Material family (supplied)	Candidate governing standards (recalled; verify)	Crystal structure at RT (qualified)	Phases & typical grain / microstructure	Property rows in Sheet 3 [calc]	Designation verification status	Reviewer note
1	AISI 1006	AISI/SAE 1006 (UNS G10060)	Low-carbon steel	ASTM A29/A29M (bar chemistry); A1008/A1011 (sheet); SAE J403	BCC α-ferrite + orthorhombic Fe3C (cementite) in pearlite	Ferrite-dominant with minor pearlite; equiaxed ferrite grains (hot-rolled/normalised) or elongated (cold-worked). Grain size set by processing — specify per ASTM E112.	0	Conditionally verified	Mostly flat-rolled; temper/finish must be stated.
2	AISI 1008	AISI/SAE 1008 (UNS G10080)	Low-carbon steel	ASTM A29/A29M; A1008/A1011; SAE J403	BCC α-ferrite + orthorhombic Fe3C (cementite) in pearlite	Ferrite-dominant with minor pearlite; equiaxed ferrite grains (hot-rolled/normalised) or elongated (cold-worked). Grain size set by processing — specify per ASTM E112.	0	Conditionally verified	Deep-drawing/sheet use.
3	AISI 1010	AISI/SAE 1010 (UNS G10100)	Low-carbon steel	ASTM A29/A29M; A108 (cold-finished)	BCC α-ferrite + orthorhombic Fe3C (cementite) in pearlite	Ferrite-dominant with minor pearlite; equiaxed ferrite grains (hot-rolled/normalised) or elongated (cold-worked). Grain size set by processing — specify per ASTM E112.	0	Conditionally verified	Case-hardenable.
4	AISI 1018	AISI/SAE 1018 (UNS G10180)	Low-carbon steel	ASTM A29/A29M; A108	BCC α-ferrite + orthorhombic Fe3C (cementite) in pearlite	Ferrite-dominant with minor pearlite; equiaxed ferrite grains (hot-rolled/normalised) or elongated (cold-worked). Grain size set by processing — specify per ASTM E112.	1	Conditionally verified	Hot-rolled vs cold-drawn properties differ substantially.
5	AISI 1020	AISI/SAE 1020 (UNS G10200)	Low-carbon steel	ASTM A29/A29M; A108	BCC α-ferrite + orthorhombic Fe3C (cementite) in pearlite	Ferrite-dominant with minor pearlite; equiaxed ferrite grains (hot-rolled/normalised) or elongated (cold-worked). Grain size set by processing — specify per ASTM E112.	0	Conditionally verified	Case-hardening response is limited vs alloy steels.
6	AISI 1025	AISI/SAE 1025 (UNS G10250)	Carbon steel	ASTM A29/A29M	BCC α-ferrite + orthorhombic Fe3C (cementite) in pearlite	Ferrite-dominant with minor pearlite; equiaxed ferrite grains (hot-rolled/normalised) or elongated (cold-worked). Grain size set by processing — specify per ASTM E112.	0	Conditionally verified	
7	AISI 1035	AISI/SAE 1035 (UNS G10350)	Medium-carbon steel	ASTM A29/A29M	BCC α-ferrite + Fe3C (pearlite); BCT martensite if quenched	Ferrite + pearlite (lever-rule estimate ≈44% pearlite at 0.35%C, ≈64% at 0.50%C); tempered martensite after Q&T. Hardenability is carbon-controlled (modest).	0	Conditionally verified	
8	AISI 1040	AISI/SAE 1040 (UNS G10400)	Medium-carbon steel	ASTM A29/A29M	BCC α-ferrite + Fe3C (pearlite); BCT martensite if quenched	Ferrite + pearlite (lever-rule estimate ≈44% pearlite at 0.35%C, ≈64% at 0.50%C); tempered martensite after Q&T. Hardenability is carbon-controlled (modest).	0	Conditionally verified	
9	AISI 1045 / C45	AISI/SAE 1045 (UNS G10450). EN C45 (1.0503) is similar, NOT identical	Medium-carbon steel	ASTM A29/A29M; EN 10083-2 (C45)	BCC α-ferrite + Fe3C (pearlite); BCT martensite if quenched	Ferrite + pearlite (lever-rule estimate ≈44% pearlite at 0.35%C, ≈64% at 0.50%C); tempered martensite after Q&T. Hardenability is carbon-controlled (modest).	1	Conditionally verified	Slash implies equivalence — see Errors E02.
10	AISI 1050	AISI/SAE 1050 (UNS G10500)	Medium-carbon steel	ASTM A29/A29M	BCC α-ferrite + Fe3C (pearlite); BCT martensite if quenched	Ferrite + pearlite (lever-rule estimate ≈44% pearlite at 0.35%C, ≈64% at 0.50%C); tempered martensite after Q&T. Hardenability is carbon-controlled (modest).	0	Conditionally verified	
11	AISI 1060	AISI/SAE 1060 (UNS G10600)	High-carbon steel	ASTM A29/A29M	BCC α-ferrite + Fe3C; BCT martensite if quenched	Mostly pearlite (~77% at 0.60%C by lever rule); fine pearlite/spheroidised or tempered martensite for springs/wear parts.	0	Conditionally verified	
12	IS 2062 E250	IS 2062:2011 E250 + quality designation (A / BR / B0 / C)	Structural steel	IS 2062 (hot-rolled structural steel); impact/test options depend on quality designation	BCC α-ferrite + orthorhombic Fe3C (cementite) in pearlite	Ferrite-dominant with minor pearlite; equiaxed ferrite grains (hot-rolled/normalised) or elongated (cold-worked). Grain size set by processing — specify per ASTM E112.	1	Conditionally verified	Grade without quality designation is incomplete; min. yield falls with thickness (E1).
13	IS 2062 E350	IS 2062:2011 E350 + quality designation	Structural steel	IS 2062	BCC ferrite + pearlite/bainite; fine Nb/V/Ti carbonitride precipitates	Fine-grained ferrite-pearlite from controlled rolling; strength from grain refinement + precipitation. Weld HAZ grain coarsening can reduce toughness.	1	Conditionally verified	As E250; check thickness bands.
14	ASTM A36	ASTM A36/A36M	Structural steel	ASTM A36/A36M	BCC α-ferrite + orthorhombic Fe3C (cementite) in pearlite	Ferrite-dominant with minor pearlite; equiaxed ferrite grains (hot-rolled/normalised) or elongated (cold-worked). Grain size set by processing — specify per ASTM E112.	1	Conditionally verified	Plate/shapes/bar; Charpy not required unless ordered.
15	ASTM A572 Grade 50	ASTM A572/A572M Grade 50	HSLA structural steel	ASTM A572/A572M	BCC ferrite + pearlite/bainite; fine Nb/V/Ti carbonitride precipitates	Fine-grained ferrite-pearlite from controlled rolling; strength from grain refinement + precipitation. Weld HAZ grain coarsening can reduce toughness.	1	Conditionally verified	Type (1–5) chemistry should be stated.
16	AISI 4130	AISI/SAE 4130 (UNS G41300)	Cr-Mo alloy steel	ASTM A29/A29M; tubing ASTM A519	Tempered martensite: BCC ferrite matrix + alloy carbides (as-quenched martensite is BCT)	Q&T gives tempered martensite/bainite; prior-austenite grain size controls toughness. Risks: temper embrittlement, quench cracking, hydrogen embrittlement at high strength.	1	Conditionally verified	Weld preheat/PWHT needed for thick sections.
17	AISI 4140 / 42CrMo4	AISI/SAE 4140 (UNS G41400). EN 42CrMo4 (1.7225) is similar, NOT identical	Cr-Mo alloy steel	ASTM A29/A29M; EN 10083-3 (42CrMo4)	Tempered martensite: BCC ferrite matrix + alloy carbides (as-quenched martensite is BCT)	Q&T gives tempered martensite/bainite; prior-austenite grain size controls toughness. Risks: temper embrittlement, quench cracking, hydrogen embrittlement at high strength.	2	Conditionally verified	Cr/Mo/Mn/C ranges differ slightly — see Errors E03.
18	AISI 4340	AISI/SAE 4340 (UNS G43400)	Ni-Cr-Mo alloy steel	ASTM A29/A29M	Tempered martensite: BCC ferrite matrix + alloy carbides (as-quenched martensite is BCT)	Q&T gives tempered martensite/bainite; prior-austenite grain size controls toughness. Risks: temper embrittlement, quench cracking, hydrogen embrittlement at high strength.	1	Conditionally verified	High hardenability; hydrogen-embrittlement sensitive at high strength.
19	AISI 8620	AISI/SAE 8620 (UNS G86200)	Case-hardening alloy steel	ASTM A29/A29M	Case: high-C tempered martensite (BCT→BCC) + carbides + retained austenite (FCC). Core: low-C martensite/bainite/ferrite	Carburised case (~0.8%C at surface) over a tough low-carbon core; retained austenite, carbide networks and grain-boundary oxidation must be controlled; grinding burn risk.	1	Conditionally verified	Gear/shaft case-hardening steel.
20	AISI 9310	AISI/SAE 9310 (UNS G93100)	Case-hardening gear steel	ASTM A29/A29M (aerospace gear grade often per AMS)	Case: high-C tempered martensite (BCT→BCC) + carbides + retained austenite (FCC). Core: low-C martensite/bainite/ferrite	Carburised case (~0.8%C at surface) over a tough low-carbon core; retained austenite, carbide networks and grain-boundary oxidation must be controlled; grinding burn risk.	0	Conditionally verified	Heavy-duty/aero gears; vacuum-melt versions exist.
21	16MnCr5	16MnCr5 (1.7131)	Case-hardening steel	EN 10084 (case-hardening steels)	Case: high-C tempered martensite (BCT→BCC) + carbides + retained austenite (FCC). Core: low-C martensite/bainite/ferrite	Carburised case (~0.8%C at surface) over a tough low-carbon core; retained austenite, carbide networks and grain-boundary oxidation must be controlled; grinding burn risk.	0	Conditionally verified	
22	20MnCr5	20MnCr5 (1.7147)	Case-hardening steel	EN 10084	Case: high-C tempered martensite (BCT→BCC) + carbides + retained austenite (FCC). Core: low-C martensite/bainite/ferrite	Carburised case (~0.8%C at surface) over a tough low-carbon core; retained austenite, carbide networks and grain-boundary oxidation must be controlled; grinding burn risk.	0	Conditionally verified	
23	40Cr	40Cr (China, GB/T 3077)	Alloy steel	GB/T 3077 (China)	Tempered martensite: BCC ferrite matrix + alloy carbides (as-quenched martensite is BCT)	Q&T gives tempered martensite/bainite; prior-austenite grain size controls toughness. Risks: temper embrittlement, quench cracking, hydrogen embrittlement at high strength.	0	Conditionally verified	Chinese designation; no BIS/ASTM equivalence asserted (approx. SAE 5140-type) — Errors E06.
24	20CrMo	20CrMo (China, GB/T 3077)	Alloy steel	GB/T 3077 (China)	Tempered martensite: BCC ferrite matrix + alloy carbides (as-quenched martensite is BCT)	Q&T gives tempered martensite/bainite; prior-austenite grain size controls toughness. Risks: temper embrittlement, quench cracking, hydrogen embrittlement at high strength.	0	Conditionally verified	Chinese designation; carburising/structural; no BIS equivalence asserted.
25	25CrMo4	25CrMo4 (1.7218)	Cr-Mo alloy steel	EN 10083-3	Tempered martensite: BCC ferrite matrix + alloy carbides (as-quenched martensite is BCT)	Q&T gives tempered martensite/bainite; prior-austenite grain size controls toughness. Risks: temper embrittlement, quench cracking, hydrogen embrittlement at high strength.	0	Conditionally verified	
26	34CrNiMo6	34CrNiMo6 (1.6582)	Ni-Cr-Mo alloy steel	EN 10083-3	Tempered martensite: BCC ferrite matrix + alloy carbides (as-quenched martensite is BCT)	Q&T gives tempered martensite/bainite; prior-austenite grain size controls toughness. Risks: temper embrittlement, quench cracking, hydrogen embrittlement at high strength.	0	Conditionally verified	Closest modern grade to legacy EN24 (not identical).
27	18CrNiMo7-6	18CrNiMo7-6 (1.6587)	Case-hardening alloy steel	EN 10084	Case: high-C tempered martensite (BCT→BCC) + carbides + retained austenite (FCC). Core: low-C martensite/bainite/ferrite	Carburised case (~0.8%C at surface) over a tough low-carbon core; retained austenite, carbide networks and grain-boundary oxidation must be controlled; grinding burn risk.	0	Conditionally verified	Heavy gears; high core toughness.
28	EN8 (verify product standard)	LEGACY: EN8 = BS 970 080M40 (withdrawn). Modern: C40/C45 (EN 10083-2)	Engineering carbon steel	BS 970 withdrawn → BS EN 10083-2	BCC α-ferrite + Fe3C (pearlite); BCT martensite if quenched	Ferrite + pearlite (lever-rule estimate ≈44% pearlite at 0.35%C, ≈64% at 0.50%C); tempered martensite after Q&T. Hardenability is carbon-controlled (modest).	0	Incorrect	Obsolete designation; modern grade similar not identical — Errors E01.
29	EN19 (verify product standard)	LEGACY: EN19 = BS 970 709M40/708M40 (withdrawn). Modern: 42CrMo4	Engineering alloy steel	BS 970 withdrawn → EN 10083-3	Tempered martensite: BCC ferrite matrix + alloy carbides (as-quenched martensite is BCT)	Q&T gives tempered martensite/bainite; prior-austenite grain size controls toughness. Risks: temper embrittlement, quench cracking, hydrogen embrittlement at high strength.	0	Incorrect	Obsolete designation — Errors E01.
30	EN24 (verify product standard)	LEGACY: EN24 = BS 970 817M40 (withdrawn). Modern: 34CrNiMo6 (approx.)	Engineering alloy steel	BS 970 withdrawn → EN 10083-3	Tempered martensite: BCC ferrite matrix + alloy carbides (as-quenched martensite is BCT)	Q&T gives tempered martensite/bainite; prior-austenite grain size controls toughness. Risks: temper embrittlement, quench cracking, hydrogen embrittlement at high strength.	0	Incorrect	Obsolete designation — Errors E01.
31	SS 201	ASTM A240 Type 201 (UNS S20100)	Stainless steel	ASTM A240/A240M; IS 6911 (plate/sheet/strip)	FCC γ-austenite	Single-phase austenite with annealing twins; strengthened by cold work (strain-induced martensite possible in 301/304). Sensitisation risk (Cr carbides at grain boundaries) if held ~450–850 °C with higher C.	0	Conditionally verified	Mn/N replace part of Ni; lower corrosion resistance; frequently mis-sold as 304 — verify by PMI.
32	SS 301	ASTM A240 Type 301 (UNS S30100)	Austenitic stainless steel	ASTM A240/A240M; A666 (cold-worked tempers)	FCC γ-austenite	Single-phase austenite with annealing twins; strengthened by cold work (strain-induced martensite possible in 301/304). Sensitisation risk (Cr carbides at grain boundaries) if held ~450–850 °C with higher C.	0	Conditionally verified	Strain-hardens rapidly; condition (annealed/¼H…) must be stated.

No.	Designation (as supplied)	Corrected / preferred designation	Material family (supplied)	Candidate governing standards (recalled; verify)	Crystal structure at RT (qualified)	Phases & typical grain / microstructure	Property rows in Sheet 3 [calc]	Designation verification status	Reviewer note
33	SS 304	ASTM A240 Type 304 (UNS S30400)	Austenitic stainless steel	ASTM A240/A240M; IS 6911	FCC γ-austenite	Single-phase austenite with annealing twins; strengthened by cold work (strain-induced martensite possible in 301/304). Sensitisation risk (Cr carbides at grain boundaries) if held ~450–850 °C with higher C.	1	Conditionally verified	
34	SS 304L	ASTM A240 Type 304L (UNS S30403)	Low-carbon austenitic stainless	ASTM A240/A240M; IS 6911	FCC γ-austenite	Single-phase austenite with annealing twins; strengthened by cold work (strain-induced martensite possible in 301/304). Sensitisation risk (Cr carbides at grain boundaries) if held ~450–850 °C with higher C.	0	Conditionally verified	Low-C for weld sensitisation resistance.
35	SS 316	ASTM A240 Type 316 (UNS S31600)	Austenitic stainless steel	ASTM A240/A240M; IS 6911	FCC γ-austenite	Single-phase austenite with annealing twins; strengthened by cold work (strain-induced martensite possible in 301/304). Sensitisation risk (Cr carbides at grain boundaries) if held ~450–850 °C with higher C.	0	Conditionally verified	
36	SS 316L	ASTM A240 Type 316L (UNS S31603)	Low-carbon austenitic stainless	ASTM A240/A240M; IS 6911	FCC γ-austenite	Single-phase austenite with annealing twins; strengthened by cold work (strain-induced martensite possible in 301/304). Sensitisation risk (Cr carbides at grain boundaries) if held ~450–850 °C with higher C.	1	Conditionally verified	
37	SS 321	ASTM A240 Type 321 (UNS S32100)	Stabilized austenitic stainless	ASTM A240/A240M	FCC γ-austenite	Single-phase austenite with annealing twins; strengthened by cold work (strain-induced martensite possible in 301/304). Sensitisation risk (Cr carbides at grain boundaries) if held ~450–850 °C with higher C.	0	Conditionally verified	Ti-stabilised.
38	SS 310S	ASTM A240 Type 310S (UNS S31008)	Heat-resistant stainless	ASTM A240/A240M	FCC γ-austenite	Single-phase austenite with annealing twins; strengthened by cold work (strain-induced martensite possible in 301/304). Sensitisation risk (Cr carbides at grain boundaries) if held ~450–850 °C with higher C.	0	Conditionally verified	High Cr-Ni for oxidation resistance; sigma risk on long exposure.
39	SS 410	ASTM A240 Type 410 (UNS S41000)	Martensitic stainless	ASTM A240/A240M; A276 (bar)	BCT martensite as-quenched → tempered martensite (BCC + carbides); annealed: ferrite + carbides	Hardenable stainless; high-C 440C contains large primary Cr carbides. Tempering window controls corrosion resistance and toughness.	0	Conditionally verified	Condition (annealed/hardened-tempered) must be stated.
40	SS 420	ASTM A240/A276 Type 420 (UNS S42000)	Martensitic stainless	ASTM A276	BCT martensite as-quenched → tempered martensite (BCC + carbides); annealed: ferrite + carbides	Hardenable stainless; high-C 440C contains large primary Cr carbides. Tempering window controls corrosion resistance and toughness.	0	Conditionally verified	
41	SS 430	ASTM A240 Type 430 (UNS S43000)	Ferritic stainless	ASTM A240/A240M	BCC α-ferrite	Single-phase ferrite (no martensitic hardening); grain coarsening in welds; 475 °C embrittlement and sigma risk at long exposure.	1	Conditionally verified	
42	SS 440C	ASTM A276 Type 440C (UNS S44004)	High-carbon martensitic stainless	ASTM A276	BCT martensite as-quenched → tempered martensite (BCC + carbides); annealed: ferrite + carbides	Hardenable stainless; high-C 440C contains large primary Cr carbides. Tempering window controls corrosion resistance and toughness.	1	Conditionally verified	Bearing/valve parts; corrosion resistance lower than 304/316.
43	Duplex SS 2205	Duplex 2205: UNS S32205 (older S31803 range) — EN 1.4462	Duplex stainless	ASTM A240/A240M; EN 10088	≈50:50 BCC δ-ferrite + FCC γ-austenite	Phase balance (≈40–60% ferrite) set by chemistry + solution anneal/quench; intermetallic (sigma, chi) risk ≈600–1000 °C; 475 °C embrittlement of ferrite.	1	Conditionally verified	Two UNS numbers exist — Errors E09.
44	SS 17-4 PH	17-4 PH: UNS S17400 (Type 630) + condition H900…H1150	Precipitation-hardening stainless	ASTM A564/A564M (bar), A693 (plate)	Martensitic (BCC/BCT) matrix + Cu-rich precipitates	Solution-treated lath martensite; ageing (H900–H1150) precipitates Cu-rich phase — condition sets strength/toughness/SCC resistance.	1	Conditionally verified	Condition mandatory for any property statement — Errors E10.
45	AISI O1	AISI O1 (UNS T31501)	Tool steel	ASTM A681	BCT martensite + alloy carbides; some retained austenite (FCC)	Hardened/tempered martensite with carbides (cementite in O1; Cr-rich M7C3 in D2). Distortion, quench cracking and carbide banding are control points.	1	Conditionally verified	
46	AISI A2	AISI A2 (UNS T30102)	Tool steel	ASTM A681	BCT martensite + alloy carbides; some retained austenite (FCC)	Hardened/tempered martensite with carbides (cementite in O1; Cr-rich M7C3 in D2). Distortion, quench cracking and carbide banding are control points.	1	Conditionally verified	
47	AISI D2	AISI D2 (UNS T30402)	Tool steel	ASTM A681	BCT martensite + alloy carbides; some retained austenite (FCC)	Hardened/tempered martensite with carbides (cementite in O1; Cr-rich M7C3 in D2). Distortion, quench cracking and carbide banding are control points.	1	Conditionally verified	Large Cr carbides; chipping risk.
48	AISI H11	AISI H11 (UNS T20811)	Hot-work tool steel	ASTM A681	Tempered martensite (BCC) + fine alloy carbides	Secondary hardening by Mo/V/Cr carbides; resists softening to ≈600 °C; thermal-fatigue (heat-checking) governs life.	0	Conditionally verified	
49	AISI H13	AISI H13 (UNS T20813)	Hot-work tool steel	ASTM A681	Tempered martensite (BCC) + fine alloy carbides	Secondary hardening by Mo/V/Cr carbides; resists softening to ≈600 °C; thermal-fatigue (heat-checking) governs life.	1	Conditionally verified	
50	AISI M2	AISI M2 (UNS T11302)	High-speed tool steel	ASTM A600	Tempered martensite + MC / M6C / M2C carbides	High carbide fraction gives red-hardness and wear resistance; multiple tempers needed to transform retained austenite.	1	Conditionally verified	
51	AISI M35	AISI M35 (UNS T11335)	High-speed tool steel	ASTM A600	Tempered martensite + MC / M6C / M2C carbides	High carbide fraction gives red-hardness and wear resistance; multiple tempers needed to transform retained austenite.	0	Conditionally verified	M2 + ≈5% Co.
52	AISI 52100 / 100Cr6	SAE 52100 (UNS G52986). EN 100Cr6 (1.3505) is close, not identical	Bearing steel	ASTM A295/A295M; ISO 683-17 (bearing steels)	Tempered martensite (BCC/BCT) + undissolved spheroidal (Fe,Cr)3C carbides + minor retained austenite	Spheroidise-annealed before machining; hardened/tempered to ≈60–66 HRC. Cleanliness (oxide/sulphide inclusions) controls rolling-contact fatigue.	1	Conditionally verified	Cleanliness grade (e.g. vacuum degassed) must be specified.
53	SAE 5160	SAE 5160 (UNS G51600)	Spring steel	ASTM A29/A29M; SAE J404	Tempered martensite or fine pearlite (BCC + Fe3C)	Oil-quenched and tempered to ≈42–52 HRC (typical practice); decarburisation and surface defects govern fatigue.	0	Conditionally verified	
54	SAE 9254	SAE 9254 (UNS G92540)	Spring steel	ASTM A29/A29M	Tempered martensite or fine pearlite (BCC + Fe3C)	Oil-quenched and tempered to ≈42–52 HRC (typical practice); decarburisation and surface defects govern fatigue.	0	Conditionally verified	
55	65Mn	65Mn (China, GB/T)	Spring steel	GB/T 1222 (China)	Tempered martensite or fine pearlite (BCC + Fe3C)	Oil-quenched and tempered to ≈42–52 HRC (typical practice); decarburisation and surface defects govern fatigue.	0	Conditionally verified	Chinese designation; not an IS grade.
56	IS 210 FG 150	IS 210:2009 FG 150	Gray cast iron	IS 210	Flake graphite (hexagonal) in ferritic/pearlitic BCC matrix	Type A flakes preferred; flakes act as internal notches (low tensile strength/ductility, high damping and compressive strength). Section-size sensitive.	1	Conditionally verified	Tensile strength on separately cast test bar (section sensitivity).
57	IS 210 FG 200	IS 210:2009 FG 200	Gray cast iron	IS 210	Flake graphite (hexagonal) in ferritic/pearlitic BCC matrix	Type A flakes preferred; flakes act as internal notches (low tensile strength/ductility, high damping and compressive strength). Section-size sensitive.	1	Conditionally verified	
58	IS 210 FG 260	IS 210:2009 FG 260	Gray cast iron	IS 210	Flake graphite (hexagonal) in ferritic/pearlitic BCC matrix	Type A flakes preferred; flakes act as internal notches (low tensile strength/ductility, high damping and compressive strength). Section-size sensitive.	1	Conditionally verified	
59	EN-GJL-200	EN-GJL-200 (EN 1561)	Gray cast iron	EN 1561	Flake graphite (hexagonal) in ferritic/pearlitic BCC matrix	Type A flakes preferred; flakes act as internal notches (low tensile strength/ductility, high damping and compressive strength). Section-size sensitive.	1	Conditionally verified	
60	EN-GJL-250	EN-GJL-250 (EN 1561)	Gray cast iron	EN 1561	Flake graphite (hexagonal) in ferritic/pearlitic BCC matrix	Type A flakes preferred; flakes act as internal notches (low tensile strength/ductility, high damping and compressive strength). Section-size sensitive.	1	Conditionally verified	
61	EN-GJS-400-15	EN-GJS-400-15 (EN 1563)	Ductile iron	EN 1563	Spheroidal graphite in ferritic/pearlitic matrix (BCC α + Fe3C)	Nodularity (commonly ≥80–90% required) and matrix ferrite/pearlite ratio set properties; ferritic grades ductile, pearlitic stronger. Mg treatment and section size control quality.	1	Conditionally verified	
62	EN-GJS-500-7	EN-GJS-500-7 (EN 1563)	Ductile iron	EN 1563	Spheroidal graphite in ferritic/pearlitic matrix (BCC α + Fe3C)	Nodularity (commonly ≥80–90% required) and matrix ferrite/pearlite ratio set properties; ferritic grades ductile, pearlitic stronger. Mg treatment and section size control quality.	1	Conditionally verified	
63	ASTM A536 65-45-12	ASTM A536 Grade 65-45-12	Ductile iron	ASTM A536	Spheroidal graphite in ferritic/pearlitic matrix (BCC α + Fe3C)	Nodularity (commonly ≥80–90% required) and matrix ferrite/pearlite ratio set properties; ferritic grades ductile, pearlitic stronger. Mg treatment and section size control quality.	1	Conditionally verified	
64	ASTM A48 Class 30	ASTM A48/A48M Class 30	Gray cast iron	ASTM A48/A48M	Flake graphite (hexagonal) in ferritic/pearlitic BCC matrix	Type A flakes preferred; flakes act as internal notches (low tensile strength/ductility, high damping and compressive strength). Section-size sensitive.	1	Conditionally verified	
65	Aluminium 1050	AA 1050 (UNS A91050) + temper	Aluminium alloy	ASTM B209/B221 (wrought forms)	FCC α-Al	Non-heat-treatable: strength from cold work (H tempers) and solid solution (Mg in 5xxx). 5xxx with high Mg can sensitise (β-Al3Mg2) after long exposure above ≈65 °C.	0	Conditionally verified	Temper required.
66	Aluminium 1100	AA 1100 (UNS A91100) + temper	Aluminium alloy	ASTM B209/B221	FCC α-Al	Non-heat-treatable: strength from cold work (H tempers) and solid solution (Mg in 5xxx). 5xxx with high Mg can sensitise (β-Al3Mg2) after long exposure above ≈65 °C.	0	Conditionally verified	
67	Aluminium 2014-T6	AA 2014-T6 (UNS A92014)	Heat-treated aluminium	ASTM B211/B247 (bar/forging); B221	FCC α-Al matrix + precipitates (θ′ Al2Cu in 2xxx; β″/β′ Mg2Si in 6xxx; η′/η MgZn2 in 7xxx)	Solution treat + age (T6): GP zones → coherent/semi-coherent precipitates. Grain-boundary precipitates/PFZ drive IGC & SCC; recrystallised vs unrecrystallised grain structure affects anisotropy and toughness.	1	Conditionally verified	Poor weldability, IGC/SCC sensitivity.

No.	Designation (as supplied)	Corrected / preferred designation	Material family (supplied)	Candidate governing standards (recalled; verify)	Crystal structure at RT (qualified)	Phases & typical grain / microstructure	Property rows in Sheet 3 [calc]	Designation verification status	Reviewer note
68	Aluminium 2024-T3	AA 2024-T3 (UNS A92024)	Heat-treated aluminium	ASTM B209	FCC α-Al matrix + precipitates (θ′ Al2Cu in 2xxx; β″/β′ Mg2Si in 6xxx; η′/η MgZn2 in 7xxx)	Solution treat + age (T6): GP zones → coherent/semi-coherent precipitates. Grain-boundary precipitates/PFZ drive IGC & SCC; recrystallised vs unrecrystallised grain structure affects anisotropy and toughness.	1	Conditionally verified	Alclad sheet common; not fusion-weldable.
69	Aluminium 3003-H14	AA 3003-H14 (UNS A93003)	Aluminium alloy	ASTM B209	FCC α-Al	Non-heat-treatable: strength from cold work (H tempers) and solid solution (Mg in 5xxx). 5xxx with high Mg can sensitise (β-Al3Mg2) after long exposure above ≈65 °C.	1	Conditionally verified	
70	Aluminium 5052-H32	AA 5052-H32 (UNS A95052)	Aluminium alloy	ASTM B209	FCC α-Al	Non-heat-treatable: strength from cold work (H tempers) and solid solution (Mg in 5xxx). 5xxx with high Mg can sensitise (β-Al3Mg2) after long exposure above ≈65 °C.	1	Conditionally verified	
71	Aluminium 5083-H111	AA 5083-H111 (UNS A95083) — EN 485-2 temper	Aluminium alloy	ASTM B209; EN 485-2	FCC α-Al	Non-heat-treatable: strength from cold work (H tempers) and solid solution (Mg in 5xxx). 5xxx with high Mg can sensitise (β-Al3Mg2) after long exposure above ≈65 °C.	1	Conditionally verified	H111 is an EN-system temper; ASTM equivalent differs (e.g. H116/H321).
72	Aluminium 6061-T6	AA 6061-T6 (UNS A96061)	Heat-treated aluminium	ASTM B209/B221/B211	FCC α-Al matrix + precipitates (θ′ Al2Cu in 2xxx; β″/β′ Mg2Si in 6xxx; η′/η MgZn2 in 7xxx)	Solution treat + age (T6): GP zones → coherent/semi-coherent precipitates. Grain-boundary precipitates/PFZ drive IGC & SCC; recrystallised vs unrecrystallised grain structure affects anisotropy and toughness.	1	Conditionally verified	Weld HAZ loses much of T6 strength.
73	Aluminium 6063-T5	AA 6063-T5 (UNS A96063)	Aluminium extrusion	ASTM B221	FCC α-Al matrix + precipitates (θ′ Al2Cu in 2xxx; β″/β′ Mg2Si in 6xxx; η′/η MgZn2 in 7xxx)	Solution treat + age (T6): GP zones → coherent/semi-coherent precipitates. Grain-boundary precipitates/PFZ drive IGC & SCC; recrystallised vs unrecrystallised grain structure affects anisotropy and toughness.	0	Conditionally verified	Extrusion alloy.
74	Aluminium 6063-T6	AA 6063-T6 (UNS A96063)	Aluminium extrusion	ASTM B221	FCC α-Al matrix + precipitates (θ′ Al2Cu in 2xxx; β″/β′ Mg2Si in 6xxx; η′/η MgZn2 in 7xxx)	Solution treat + age (T6): GP zones → coherent/semi-coherent precipitates. Grain-boundary precipitates/PFZ drive IGC & SCC; recrystallised vs unrecrystallised grain structure affects anisotropy and toughness.	1	Conditionally verified	
75	Aluminium 6082-T6	EN AW-6082-T6 (UNS A96082)	Heat-treated aluminium	EN 485/755	FCC α-Al matrix + precipitates (θ′ Al2Cu in 2xxx; β″/β′ Mg2Si in 6xxx; η′/η MgZn2 in 7xxx)	Solution treat + age (T6): GP zones → coherent/semi-coherent precipitates. Grain-boundary precipitates/PFZ drive IGC & SCC; recrystallised vs unrecrystallised grain structure affects anisotropy and toughness.	0	Conditionally verified	European structural 6xxx; not interchangeable with 6061 by default.
76	Aluminium 7075-T6	AA 7075-T6 (UNS A97075)	High-strength aluminium	ASTM B209/B211	FCC α-Al matrix + precipitates (θ′ Al2Cu in 2xxx; β″/β′ Mg2Si in 6xxx; η′/η MgZn2 in 7xxx)	Solution treat + age (T6): GP zones → coherent/semi-coherent precipitates. Grain-boundary precipitates/PFZ drive IGC & SCC; recrystallised vs unrecrystallised grain structure affects anisotropy and toughness.	1	Conditionally verified	SCC-susceptible in short-transverse; not fusion-weldable.
77	Aluminium A356-T6	A356.0-T6 (UNS A13560)	Cast aluminium	ASTM B26/B26M (castings)	FCC α-Al dendrites + eutectic Si (diamond cubic) + Fe/Mg intermetallics	Dendrite arm spacing, Sr/Na modification of Si, porosity and Fe-intermetallics control strength/ductility/fatigue; die casting gives fine skin + porous core.	1	Conditionally verified	Casting method and test-bar vs casting properties differ.
78	Aluminium A380	A380.0 die casting (UNS A03800)	Die-cast aluminium	ASTM B85/B85M	FCC α-Al dendrites + eutectic Si (diamond cubic) + Fe/Mg intermetallics	Dendrite arm spacing, Sr/Na modification of Si, porosity and Fe-intermetallics control strength/ductility/fatigue; die casting gives fine skin + porous core.	1	Conditionally verified	
79	Copper C10100	UNS C10100 (OFE copper)	Copper	ASTM B152 / B187	FCC Cu	Single-phase FCC; annealing twins; strength set by cold work; oxygen-free (C10100) vs ETP (C11000, Cu2O particles — hydrogen embrittlement if annealed in H2).	0	Conditionally verified	
80	Copper C11000	UNS C11000 (ETP copper)	Copper	ASTM B152 / B187	FCC Cu	Single-phase FCC; annealing twins; strength set by cold work; oxygen-free (C10100) vs ETP (C11000, Cu2O particles — hydrogen embrittlement if annealed in H2).	1	Conditionally verified	
81	Brass C26000	UNS C26000 (cartridge brass)	Brass	ASTM B36/B36M (plate/sheet)	FCC α-brass (Cu-Zn solid solution)	Single-phase α (Zn ≲35%); high cold-work response; season cracking (residual stress + ammonia) and dezincification risk in some waters.	1	Conditionally verified	Temper (O/H) required.
82	Brass C36000	UNS C36000 (free-cutting brass)	Free-machining brass	ASTM B16/B16M (rod)	FCC α + BCC β′ brass with Pb globules	Two-phase α+β′ gives hot workability; Pb globules give free-cutting. Pb is restricted in some regulations (RoHS/REACH/drinking water).	0	Conditionally verified	Leaded; regulatory restrictions in some uses.
83	Bronze C51000	UNS C51000 (phosphor bronze 5% A)	Phosphor bronze	ASTM B103/B139	FCC α-Cu(Sn) solid solution (+ δ Cu31Sn8 / Pb globules in cast bearing bronze)	Wrought phosphor bronze: cold-work + Sn strengthening. Cast leaded tin bronze: α dendrites + δ + Pb islands give embeddability/lubricity.	0	Conditionally verified	
84	Bronze C93200	UNS C93200 (SAE 660 bearing bronze)	Bearing bronze	ASTM B584 (sand castings)	FCC α-Cu(Sn) solid solution (+ δ Cu31Sn8 / Pb globules in cast bearing bronze)	Wrought phosphor bronze: cold-work + Sn strengthening. Cast leaded tin bronze: α dendrites + δ + Pb islands give embeddability/lubricity.	1	Conditionally verified	Leaded.
85	Aluminium bronze C95400	UNS C95400 (aluminium bronze)	Aluminium bronze	ASTM B148	FCC α-Cu(Al) + κ intermetallic phases (Fe/Ni/Al-rich)	κ phases give strength/wear resistance and seawater resistance; slow cooling can cause dealuminification-prone phases.	1	Conditionally verified	
86	Manganese bronze C86300	UNS C86300 (manganese bronze)	Copper alloy	ASTM B584	FCC α + BCC β (high-strength brass)	Two-phase brass strengthened by Mn/Al/Fe; good strength and marine use; dezincification/selective leaching must be considered.	0	Conditionally verified	
87	Titanium Grade 2	Ti Grade 2 (UNS R50400)	Commercially pure titanium	ASTM B265/B348/B381	HCP α-Ti	Single-phase α; O/Fe interstitials control strength by grade (Gr 1–4). Twinning + prismatic slip; hydrogen pickup embrittles.	1	Conditionally verified	
88	Ti-6Al-4V Grade 5	Ti-6Al-4V Grade 5 (UNS R56400); ELI = Grade 23 (R56407)	Titanium alloy	ASTM B265/B348/B381; F136 (ELI implant)	HCP α + BCC β (two-phase)	Equiaxed (mill-annealed) vs lamellar/Widmanstätten (β-processed) vs bimodal: trade-off between strength/ductility/fatigue initiation and fracture toughness/crack growth. β transus ≈995 °C (typical).	1	Conditionally verified	Condition (annealed/STA/β-processed) required.
89	Magnesium AZ31B	AZ31B (UNS M11311)	Magnesium alloy	ASTM B90/B90M	HCP α-Mg (+ β-Mg17Al12 in AZ91D)	Limited RT slip systems (basal-dominant) → anisotropy and low RT formability; die-cast AZ91D has fine skin + porous core; poor creep >≈120 °C.	1	Conditionally verified	Flammability of fines/chips.
90	Magnesium AZ91D	AZ91D (UNS M11916)	Cast magnesium	ASTM B94	HCP α-Mg (+ β-Mg17Al12 in AZ91D)	Limited RT slip systems (basal-dominant) → anisotropy and low RT formability; die-cast AZ91D has fine skin + porous core; poor creep >≈120 °C.	1	Conditionally verified	Die-cast; creep limited.
91	Inconel 600	UNS N06600 (Inconel® 600 — trademark)	Nickel alloy	ASTM B166/B168	FCC γ (single-phase, solid-solution strengthened)	Single-phase FCC with twins; carbides at grain boundaries (e.g. in 600/625); strengthening from Mo/Nb/Cr/Fe in solution; sensitisation/precipitation (μ, P phases) on long exposure.	1	Conditionally verified	Trade name → use UNS.
92	Inconel 625	UNS N06625 (Inconel® 625 — trademark)	Nickel alloy	ASTM B443/B446	FCC γ (single-phase, solid-solution strengthened)	Single-phase FCC with twins; carbides at grain boundaries (e.g. in 600/625); strengthening from Mo/Nb/Cr/Fe in solution; sensitisation/precipitation (μ, P phases) on long exposure.	1	Conditionally verified	Grade 1 (annealed) vs Grade 2 (solution annealed).
93	Inconel 718	UNS N07718 (Inconel® 718 — trademark)	Precipitation-hardening nickel alloy	ASTM B637; AMS aerospace specs	FCC γ matrix + γ″ (Ni3Nb, BCT D0_22) + γ′ (Ni3(Al,Ti), L1_2) + δ (orthorhombic)	Solution + double ageing precipitates γ″ (main strengthener, ≲650 °C service); δ-phase at grain boundaries controls grain size/notch behaviour; Laves phase from segregation.	1	Conditionally verified	Heat-treat condition required.
94	Monel 400	UNS N04400 (Monel® 400 — trademark)	Nickel-copper alloy	ASTM B127/B164	FCC γ (single-phase, solid-solution strengthened)	Single-phase FCC with twins; carbides at grain boundaries (e.g. in 600/625); strengthening from Mo/Nb/Cr/Fe in solution; sensitisation/precipitation (μ, P phases) on long exposure.	1	Conditionally verified	
95	Hastelloy C-276	UNS N10276 (Hastelloy® C-276 — trademark)	Nickel alloy	ASTM B574/B575	FCC γ (single-phase, solid-solution strengthened)	Single-phase FCC with twins; carbides at grain boundaries (e.g. in 600/625); strengthening from Mo/Nb/Cr/Fe in solution; sensitisation/precipitation (μ, P phases) on long exposure.	1	Conditionally verified	
96	Zamak 3	Zamak 3 = ASTM B86 alloy AG40A	Zinc die-casting alloy	ASTM B86 (zinc die casting alloys)	HCP η-Zn (Zn-rich) + Al-rich (FCC) phase	Die-cast fine-grained structure; creep and dimensional ageing at RT/elevated T; low melting point.	1	Conditionally verified	
97	Zamak 5	Zamak 5 = ASTM B86 alloy AC41A	Zinc die-casting alloy	ASTM B86	HCP η-Zn (Zn-rich) + Al-rich (FCC) phase	Die-cast fine-grained structure; creep and dimensional ageing at RT/elevated T; low melting point.	1	Conditionally verified	Higher Cu than Zamak 3.
98	PA6	PA6 (nylon 6) — grade/filler/moisture state required	Engineering polymer	ASTM D4066 (nylon materials); manufacturer datasheet	Semi-crystalline polymer: crystalline lamellae + amorphous regions (no metallic lattice)	Crystallinity and spherulite size depend on cooling rate/nucleation; Tg and Tm define service limits; moisture (PA) and creep affect stiffness.	1	Unverified	Properties are datasheet/grade specific.
99	PA66	PA66 (nylon 66) — grade/filler/moisture state required	Engineering polymer	ASTM D4066; datasheet	Semi-crystalline polymer: crystalline lamellae + amorphous regions (no metallic lattice)	Crystallinity and spherulite size depend on cooling rate/nucleation; Tg and Tm define service limits; moisture (PA) and creep affect stiffness.	1	Unverified	
100	POM-C	POM-C (acetal copolymer)	Acetal copolymer	ASTM D6778 (POM materials); datasheet	Semi-crystalline polymer: crystalline lamellae + amorphous regions (no metallic lattice)	Crystallinity and spherulite size depend on cooling rate/nucleation; Tg and Tm define service limits; moisture (PA) and creep affect stiffness.	1	Unverified	
101	POM-H	POM-H (acetal homopolymer)	Acetal homopolymer	ASTM D6778; datasheet	Semi-crystalline polymer: crystalline lamellae + amorphous regions (no metallic lattice)	Crystallinity and spherulite size depend on cooling rate/nucleation; Tg and Tm define service limits; moisture (PA) and creep affect stiffness.	1	Unverified	
102	PTFE	PTFE	Fluoropolymer	ASTM D4894/D4895 (PTFE resins) — verify; datasheet	Semi-crystalline polymer: crystalline lamellae + amorphous regions (no metallic lattice)	Crystallinity and spherulite size depend on cooling rate/nucleation; Tg and Tm define service limits; moisture (PA) and creep affect stiffness.	1	Unverified	Cold flow.

No.	Designation (as supplied)	Corrected / preferred designation	Material family (supplied)	Candidate governing standards (recalled; verify)	Crystal structure at RT (qualified)	Phases & typical grain / microstructure	Property rows in Sheet 3 [calc]	Designation verification status	Reviewer note
103	PEEK	PEEK (unfilled vs filled must be stated)	High-performance polymer	Manufacturer datasheet	Semi-crystalline polymer: crystalline lamellae + amorphous regions (no metallic lattice)	Crystallinity and spherulite size depend on cooling rate/nucleation; Tg and Tm define service limits; moisture (PA) and creep affect stiffness.	1	Unverified	
104	ABS	ABS	Engineering polymer	ASTM D4673 (ABS) — verify; datasheet	Amorphous polymer (no long-range order)	Glassy below Tg; stress-cracking/solvent sensitivity; molecular orientation from moulding gives anisotropy.	1	Unverified	
105	Polycarbonate (PC)	Polycarbonate (PC)	Engineering polymer	ASTM D3935 (PC) — verify; datasheet	Amorphous polymer (no long-range order)	Glassy below Tg; stress-cracking/solvent sensitivity; molecular orientation from moulding gives anisotropy.	1	Unverified	
106	UHMWPE	UHMWPE	Engineering polymer	ASTM D4020 (UHMWPE moulding/extrusion)	Semi-crystalline polymer: crystalline lamellae + amorphous regions (no metallic lattice)	Crystallinity and spherulite size depend on cooling rate/nucleation; Tg and Tm define service limits; moisture (PA) and creep affect stiffness.	1	Unverified	
107	HDPE	HDPE	Thermoplastic polymer	ASTM D4976 (PE plastics) — verify; datasheet	Semi-crystalline polymer: crystalline lamellae + amorphous regions (no metallic lattice)	Crystallinity and spherulite size depend on cooling rate/nucleation; Tg and Tm define service limits; moisture (PA) and creep affect stiffness.	0	Unverified	
108	PVC-U	PVC-U	Thermoplastic polymer	ASTM D1784 (rigid PVC compounds) — verify	Amorphous polymer (no long-range order)	Glassy below Tg; stress-cracking/solvent sensitivity; molecular orientation from moulding gives anisotropy.	0	Unverified	
109	TPU	TPU (polyester vs polyether grade, Shore hardness required)	Elastomeric polymer	Manufacturer datasheet	Segmented block elastomer: hard (crystalline) + soft domains	Phase-separated hard/soft segments give elasticity, abrasion resistance; hydrolysis/temperature sensitivity depends on polyol type.	0	Unverified	Under-specified.
110	PPS	PPS	High-performance polymer	Manufacturer datasheet	Semi-crystalline polymer: crystalline lamellae + amorphous regions (no metallic lattice)	Crystallinity and spherulite size depend on cooling rate/nucleation; Tg and Tm define service limits; moisture (PA) and creep affect stiffness.	0	Unverified	
111	PET-P	PET-P (ambiguous: semi-crystalline PET, unfilled?)	Engineering polymer	Manufacturer datasheet	Semi-crystalline polymer: crystalline lamellae + amorphous regions (no metallic lattice)	Crystallinity and spherulite size depend on cooling rate/nucleation; Tg and Tm define service limits; moisture (PA) and creep affect stiffness.	0	Unverified	Under-specified — Errors E08.
112	Alumina (Al2O3)	Alumina Al2O3 — purity (e.g. 96% / 99.5%) required	Technical ceramic	Manufacturer datasheet; test: ASTM C1161	α-Al2O3 corundum (trigonal/hexagonal)	Polycrystalline, grain size ≈1–10 µm typical; purity (96% vs 99.5%) and glassy grain-boundary phase affect strength/wear.	1	Unverified	Under-specified grade.
113	Silicon carbide (SiC)	SiC — sintered / reaction-bonded / hot-pressed required	Technical ceramic	Manufacturer datasheet; test: ASTM C1161	SiC polytypes: α (hexagonal, e.g. 6H) / β (cubic 3C)	Sintered (SSiC) vs reaction-bonded (free Si) vs hot-pressed: different strength, corrosion and temperature limits.	1	Unverified	Under-specified grade.
114	Zirconia (ZrO2)	ZrO2 — stabiliser (Y-TZP / Mg-PSZ) required	Technical ceramic	ISO 13356 (Y-TZP implants) for medical only; datasheet	ZrO2: monoclinic (RT, unstabilised), tetragonal (Y-TZP, metastable), cubic	Stabiliser (Y2O3, MgO, CeO2) controls t→m transformation toughening; low-temperature degradation (hydrothermal ageing) in Y-TZP.	1	Unverified	Under-specified grade.
115	Silicon nitride (Si3N4)	Si3N4 — sinter route required	Technical ceramic	Manufacturer datasheet; test: ASTM C1161	Si3N4: β (hexagonal) mainly, α (trigonal)	Elongated β grains (self-reinforced) with glassy boundary phase give high toughness among ceramics; sinter route (GPS, HIP, SRBSN) sets properties.	1	Unverified	Under-specified grade.
116	WC-Co cemented carbide	WC-Co hardmetal — Co wt% & WC grain size required	Hardmetal	ISO 513 classification; ASTM B311/B406/B294 test methods	WC (hexagonal) grains + Co binder (FCC/HCP)	Grain size (nano→coarse) and Co wt% set hardness–toughness trade-off; binder leaching in corrosive media.	1	Unverified	
117	CBN	CBN tool material (PCBN) — binder/CBN % required	Cutting-tool material	ISO 513 (classification of hard cutting materials)	cBN (zinc-blende cubic) in ceramic/metal binder	Polycrystalline CBN compact; binder type (TiN/ceramic vs metallic) sets toughness/thermal stability.	0	Unverified	Not a standardised material.
118	PCD	PCD tool material — diamond grain size required	Cutting-tool material	ISO 513	Diamond cubic grains sintered with Co binder	Sintered diamond compact on WC substrate; graphitises/oxidises at elevated T; unsuitable for machining ferrous metals (carbon diffusion).	0	Unverified	Not a standardised material.
119	CFRP	CFRP — fibre, matrix, layup, Vf required	Fibre composite	Test: ASTM D3039/D7264/D2344; material per prepreg datasheet	Composite: fibre (carbon: turbostratic graphitic; glass: amorphous; aramid: oriented polymer) in polymer matrix	Anisotropic laminate; properties depend on layup, fibre volume fraction, voids and cure; matrix Tg limits temperature.	1	Unverified	Class of materials, not a grade — Errors E08.
120	GFRP	GFRP — glass type, matrix, layup required	Fibre composite	Test: ASTM D3039/D7264	Composite: fibre (carbon: turbostratic graphitic; glass: amorphous; aramid: oriented polymer) in polymer matrix	Anisotropic laminate; properties depend on layup, fibre volume fraction, voids and cure; matrix Tg limits temperature.	1	Unverified	Class of materials.
121	Aramid-fibre composite	Aramid-fibre composite — fibre, matrix, layup required	Fibre composite	Datasheet	Composite: fibre (carbon: turbostratic graphitic; glass: amorphous; aramid: oriented polymer) in polymer matrix	Anisotropic laminate; properties depend on layup, fibre volume fraction, voids and cure; matrix Tg limits temperature.	0	Unverified	Class of materials.
122	NiTi / Nitinol	NiTi (Nitinol®): ASTM F2063 (medical wrought)	Shape-memory alloy	ASTM F2063; F2004 (transformation temp.)	NiTi: B2 cubic austenite ↔ B19′ monoclinic martensite	Transformation temperatures (Af, Ms) very sensitive to composition (≈±0.1 at% Ni) and thermomechanical history; superelastic vs shape-memory behaviour.	0	Conditionally verified	Af must be stated.
123	CuBe	Beryllium copper UNS C17200 (≈1.8–2.0% Be); low-Be C17510 also exists	Beryllium copper	ASTM B196/B194	FCC Cu matrix + γ″/γ′ CuBe precipitates after ageing	Solution anneal + age (e.g. AT/TH condition) gives very high strength with good conductivity; Be dust/fume is a serious health hazard.	0	Conditionally verified	Underspecified in list; Be hazard.
124	Cobalt-chromium alloy	UNDERSPECIFIED: ASTM F75 (cast CoCrMo) / F1537 (wrought CoCrMo) / F90 (L-605) / Stellite	Special alloy	ASTM F75/F1537/F90	Mixed FCC (γ) / HCP (ε) Co-Cr matrix + M23C6 carbides (grade/process dependent)	Cast vs wrought vs AM structures differ markedly; strain-induced γ→ε transformation affects wear and fatigue.	0	Unverified	Cannot assign properties without exact grade — Errors E08.
125	Invar 36	Invar 36 (UNS K93600)	Low-expansion alloy	ASTM F1684	FCC γ Fe-Ni (≈36% Ni)	Low CTE from magnetovolume effect below Curie point; cold work and temperature raise CTE; dimensional drift (creep/ageing) needs stabilising anneal.	1	Conditionally verified	
126	Kovar	Kovar (UNS K94610)	Controlled-expansion alloy	ASTM F15	FCC γ Fe-Ni-Co	Chemistry tuned so CTE matches borosilicate glass/ceramics; hydrogen-anneal and oxide layer needed for sealing.	1	Conditionally verified	
127	Molybdenum	Molybdenum (unalloyed)	Refractory metal	ASTM B386 (Mo mill products)	BCC (W, Mo)	Powder-metallurgy/worked structures; ductile-to-brittle transition near/above RT; oxidise rapidly in air at high T (Mo → volatile MoO3).	1	Conditionally verified	
128	Tungsten	Tungsten (unalloyed)	Refractory metal	ASTM B760 (plate/sheet)	BCC (W, Mo)	Powder-metallurgy/worked structures; ductile-to-brittle transition near/above RT; oxidise rapidly in air at high T (Mo → volatile MoO3).	1	Conditionally verified	
129	Graphite	UNDERSPECIFIED: graphite — isostatic/extruded/moulded grade required	Carbon material	Manufacturer datasheet	Hexagonal graphite (sp² layers) in polycrystalline filler/binder carbon	Grain size/porosity/anisotropy depend on forming route (isostatic vs extruded vs moulded); oxidises in air above ≈400–500 °C.	0	Unverified	Properties vary several-fold between grades.
130	Hadfield manganese steel	Austenitic manganese steel — ASTM A128/A128M Grade A–F	Austenitic manganese steel	ASTM A128/A128M; Indian equivalent not verified	FCC austenite (≈11–14% Mn, ≈1.0–1.4% C) + carbides in as-cast	Water-toughening dissolves carbides; strain-induced twinning/work-hardening raises surface hardness under impact; poor at low impact or with carbide networks.	1	Conditionally verified	Grade/composition required.
131	Hardox 400	Hardox® 400 (SSAB proprietary brand)	Wear-resistant steel plate	SSAB datasheet (no public product standard)	Martensite/bainite (quenched or Q&T) — BCC/BCT	Wear plate hardness from fine martensitic structure; welding heat input/preheat limits and HAZ softening matter; bend radius limits apply.	1	Conditionally verified	Property guarantees come from manufacturer.
132	Hardox 450	Hardox® 450 (SSAB proprietary brand)	Wear-resistant steel plate	SSAB datasheet	Martensite/bainite (quenched or Q&T) — BCC/BCT	Wear plate hardness from fine martensitic structure; welding heat input/preheat limits and HAZ softening matter; bend radius limits apply.	1	Conditionally verified	As Hardox 400.
Reviewer-suggested gap list — NOT in the upload and NOT counted in coverage									
G1	904L (UNS N08904)	UNS N08904	Super-austenitic stainless	ASTM A240	FCC γ-austenite	Single-phase austenite with annealing twins; strengthened by cold work (strain-induced martensite possible in 301/304). Sensitisation risk (Cr carbides at grain boundaries) if held ~450–850 °C with higher C.		Not applicable (gap list)	Pitting/chloride service.
G2	2507 super duplex (UNS S32750)	UNS S32750	Super-duplex stainless	ASTM A240	≈50:50 BCC δ-ferrite + FCC γ-austenite	Phase balance (≈40–60% ferrite) set by chemistry + solution anneal/quench; intermetallic (sigma, chi) risk ≈600–1000 °C; 475 °C embrittlement of ferrite.		Not applicable (gap list)	Higher PREN than 2205.
G3	15-5 PH (UNS S15500)	UNS S15500	PH stainless	ASTM A564	Martensitic (BCC/BCT) matrix + Cu-rich precipitates	Solution-treated lath martensite; ageing (H900–H1150) precipitates Cu-rich phase — condition sets strength/toughness/SCC resistance.		Not applicable (gap list)	Better transverse toughness than 17-4.
G4	Maraging steel (18Ni 250/300)	Grade 250/300	Ultra-high-strength steel	AMS/ASTM A579 (verify)	Tempered martensite: BCC ferrite matrix + alloy carbides (as-quenched martensite is BCT)	Q&T gives tempered martensite/bainite; prior-austenite grain size controls toughness. Risks: temper embrittlement, quench cracking, hydrogen embrittlement at high strength.		Not applicable (gap list)	Tools/aerospace shafts.
G5	IS 1570 / IS 5517 Indian wrought steels (e.g. 45C8, 40Cr1…)	IS 1570 series	Indian carbon & alloy steels	IS 1570, IS 5517 (verify)	BCC α-ferrite + Fe3C (pearlite); BCT martensite if quenched	Ferrite + pearlite (lever-rule estimate ≈44% pearlite at 0.35%C, ≈64% at 0.50%C); tempered martensite after Q&T. Hardenability is carbon-controlled (modest).		Not applicable (gap list)	Needed for India-native procurement.
G6	S355 (EN 10025-2) / S690QL (EN 10025-6)	EN 10025	Structural steel	EN 10025-2/-6	BCC ferrite + pearlite/bainite; fine Nb/V/Ti carbonitride precipitates	Fine-grained ferrite-pearlite from controlled rolling; strength from grain refinement + precipitation. Weld HAZ grain coarsening can reduce toughness.		Not applicable (gap list)	Common EU structural grades.
G7	AlSi10Mg (L-PBF)	ASTM F3318	AM aluminium	ASTM F3318 (verify)	Depends on alloy; AM structures are non-equilibrium (columnar grains along build direction, fine cellular microstructure)	Anisotropy, porosity/lack-of-fusion, residual stress and surface roughness control performance; HIP/stress-relief commonly needed.		Not applicable (gap list)	Heat-treat/HIP state required.
G8	316L (L-PBF)	ASTM F3184	AM stainless	ASTM F3184 (verify)	Depends on alloy; AM structures are non-equilibrium (columnar grains along build direction, fine cellular microstructure)	Anisotropy, porosity/lack-of-fusion, residual stress and surface roughness control performance; HIP/stress-relief commonly needed.		Not applicable (gap list)	
G9	Ti-6Al-4V / ELI (L-PBF)	ASTM F2924/F3001	AM titanium	ASTM F2924, F3001 (verify)	Depends on alloy; AM structures are non-equilibrium (columnar grains along build direction, fine cellular microstructure)	Anisotropy, porosity/lack-of-fusion, residual stress and surface roughness control performance; HIP/stress-relief commonly needed.		Not applicable (gap list)	HIP common for fatigue-critical.

No.	Designation (as supplied)	Corrected / preferred designation	Material family (supplied)	Candidate governing standards (recalled; verify)	Crystal structure at RT (qualified)	Phases & typical grain / microstructure	Property rows in Sheet 3 [calc]	Designation verification status	Reviewer note
G10	Inconel 718 (L-PBF)	ASTM F3055	AM nickel superalloy	ASTM F3055 (verify)	Depends on alloy; AM structures are non-equilibrium (columnar grains along build direction, fine cellular microstructure)	Anisotropy, porosity/lack-of-fusion, residual stress and surface roughness control performance; HIP/stress-relief commonly needed.		Not applicable (gap list)	
G11	Scalmalloy® (Al-Mg-Sc)	Proprietary	AM aluminium (proprietary)	Manufacturer	Depends on alloy; AM structures are non-equilibrium (columnar grains along build direction, fine cellular microstructure)	Anisotropy, porosity/lack-of-fusion, residual stress and surface roughness control performance; HIP/stress-relief commonly needed.		Not applicable (gap list)	
G12	CuCrZr (UNS C18150)	UNS C18150	Copper alloy	ASTM B (verify)	FCC Cu	Single-phase FCC; annealing twins; strength set by cold work; oxygen-free (C10100) vs ETP (C11000, Cu2O particles — hydrogen embrittlement if annealed in H2).		Not applicable (gap list)	Resistance-welding electrodes, heat sinks.
G13	Austempered ductile iron (ADI)	ASTM A897	Cast iron	ASTM A897 (verify)	Spheroidal graphite in ferritic/pearlitic matrix (BCC α + Fe3C)	Nodularity (commonly ≥80–90% required) and matrix ferrite/pearlite ratio set properties; ferritic grades ductile, pearlitic stronger. Mg treatment and section size control quality.		Not applicable (gap list)	Gear/wear parts.
G14	Cast steel (IS 1030 / ASTM A27)	IS 1030 / A27	Cast steel	IS 1030, ASTM A27 (verify)	BCC α-ferrite + Fe3C (pearlite); BCT martensite if quenched	Ferrite + pearlite (lever-rule estimate ≈44% pearlite at 0.35%C, ≈64% at 0.50%C); tempered martensite after Q&T. Hardenability is carbon-controlled (modest).		Not applicable (gap list)	Cast structural parts.
G15	PA12 / PA11 (powder-bed fusion)	Datasheet	AM polymer	Manufacturer	Semi-crystalline polymer: crystalline lamellae + amorphous regions (no metallic lattice)	Crystallinity and spherulite size depend on cooling rate/nucleation; Tg and Tm define service limits; moisture (PA) and creep affect stiffness.		Not applicable (gap list)	Robot enclosures/gears.
G16	Weathering steel (ASTM A588)	ASTM A588	HSLA steel	ASTM A588 (verify)	BCC ferrite + pearlite/bainite; fine Nb/V/Ti carbonitride precipitates	Fine-grained ferrite-pearlite from controlled rolling; strength from grain refinement + precipitation. Weld HAZ grain coarsening can reduce toughness.		Not applicable (gap list)	Atmospheric corrosion.

Sheet 3 — Material Properties (one row per grade / condition / product form)

Blue = sourced value (recalled, see Sources); grey = Excel formula; no values were invented for blank cells. All RT unless stated.

Prop ID	Master No.	Designation	Condition / product form	Value basis	Density ρ (kg/m³)	Density (g/cm³) [calc]	Young's modulus E (GPa)	Poisson ν	modulus G (GPa) [derived,	Yield / proof strength (MPa)	Ultimate tensile strength (MPa)	Flexural strength (MPa)	Elongation at fracture (%)	Hardness (as stated)	Thermal conductivity k (W/(m·K))	CTE (µm/(m·K))	Electrical resistivity (Ω·m)	Strength used [calc]	Specific strength (kN·m/kg) [calc]	Specific stiffness (MN·m/kg) [calc]	Impact / fatigue / KIC / other	Test temperature	Source class	Confidence	Verification status	Limitations / notes
P01	4	AISI 1018	Cold-drawn bar	Typical	7,870	7.870	205.0	0.290	79.5	370	440		15	≈126 HBW (handbook)	51.9	11.9	1.59E-07	UTS	55.9	26.05		RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	Hot-rolled condition is lower strength; MatWeb-type values.
P02	9	AISI 1045	Hot-rolled bar	Typical	7,850	7.850	205.0	0.290	79.5	310	565		16	≈163 HBW	49.8	11.9		UTS	72.0	26.11		RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	Cold-drawn and Q&T conditions differ strongly.
P03	14	ASTM A36	Plate/shapes (specified)	Specified minimum	7,850	7.850	200.0	0.300	76.9	250	400		20	n/a	51.9	11.7		UTS	51.0	25.48	UTS spec range 400–550; elong. 20% (200 mm GL) / 23% (50 mm GL)	RT	Product-standard value; recalled, standard text not re-read	Medium	Conditionally verified	Minimum values; ρ, E, k, CTE are generic carbon-steel handbook values.
P04	12	IS 2062 E250	Plate/section ≤20 mm, quality designation A (verify)	Specified minimum	7,850	7.850	200.0	0.300	76.9	250	410		23	n/a				UTS	52.2	25.48		RT	Product-standard value; recalled, standard text not re-read	Medium	Conditionally verified	Yield minimum drops with thickness; ρ and E are generic carbon-steel values, not IS 2062-specific.
P05	13	IS 2062 E350	Plate/section ≤20 mm (verify)	Specified minimum	7,850	7.850	200.0	0.300	76.9	350	490		22	n/a				UTS	62.4	25.48		RT	Product-standard value; recalled, standard text not re-read	Medium	Conditionally verified	As E250; confirm thickness bands in standard.
P06	15	ASTM A572 Gr 50	Plate (specified)	Specified minimum	7,850	7.850	200.0	0.300	76.9	345	450		18	n/a				UTS	57.3	25.48	Elong. 18% (200 mm GL) / 21% (50 mm GL)	RT	Product-standard value; recalled, standard text not re-read	Medium	Conditionally verified	Minimum values.
P07	16	AISI 4130	Normalised 870 °C	Typical	7,850	7.850	205.0	0.290	79.5	435	670		25.5	≈197 HBW	42.7	12.3		UTS	85.4	26.11		RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	Strongly condition-dependent (Q&T gives far higher strength).
P08	17	AISI 4140	Annealed 815 °C, furnace-cooled	Typical	7,850	7.850	205.0	0.290	79.5	417	655		25.7	≈197 HBW	42.6	12.3		UTS	83.4	26.11		RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	
P09	17	AISI 4140	Q&T (oil quench, ≈650 °C temper)	Typical	7,850	7.850	205.0	0.290	79.5	655	758		25	≈220–235 HBW	42.6	12.3		UTS	96.6	26.11	Lower tempering temperature → much higher strength, lower toughness	RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Low	Unverified	Single tempering point only; use heat-treat curves for design.
P10	18	AISI 4340	Annealed	Typical	7,850	7.850	205.0	0.290	79.5	472	745		22	≈217 HBW	44.5	12.3		UTS	94.9	26.11		RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	Q&T 4340 reaches >1000 MPa UTS — not recorded here.
P11	19	AISI 8620	Normalised 925 °C (uncarburised core)	Typical	7,850	7.850	205.0	0.290	79.5	357	632		26	≈183 HBW (core)	46.6			UTS	80.5	26.11	Carburised case typically ≈58–62 HRC (process result, not base property)	RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	
P12	52	AISI 52100	Hardened & tempered (≈150–180 °C)	Typical	7,810	7.810	201.0	0.300	77.3					≈60–66 HRC (bearing practice)				—		25.74	Rolling-contact fatigue life depends on cleanliness	RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	Strength not recorded (hardness is the control parameter).
P13	33	SS 304 (A240 Type 304)	Annealed plate/sheet (specified)	Specified minimum	8,000	8.000	193.0	0.290	74.8	205	515		40	≤201 HBW / ≤92 HRB (A240 max)	16.2	17.2	7.20E-07	UTS	64.4	24.13		RT	Product-standard value; recalled, standard text not re-read	Medium	Conditionally verified	k at 100 °C, CTE 0–100 °C. Typical YS ≈215–290 MPa.
P14	36	SS 316L (A240 Type 316L)	Annealed plate/sheet (specified)	Specified minimum	8,000	8.000	193.0	0.300	74.2	170	485		40	≤217 HBW / ≤95 HRB (A240 max)	16.3	16	7.40E-07	UTS	60.6	24.13		RT	Product-standard value; recalled, standard text not re-read	Medium	Conditionally verified	
P15	43	Duplex 2205 (UNS S32205)	Annealed plate (specified)	Specified minimum	7,800	7.800	200.0	0.300	76.9	450	655		25	≤293 HBW / ≤31 HRC (A240 max)	19	13	8.00E-07	UTS	84.0	25.64		RT	Product-standard value; recalled, standard text not re-read	Medium	Conditionally verified	Service temperature typically limited (≈250–300 °C) because of 475 °C embrittlement.
P16	44	17-4 PH (UNS S17400)	Condition H1025 (A564 min)	Specified minimum	7,780	7.780	197.0	0.270	77.6	1000	1070		12	≥35 HRC				UTS	137.5	25.32		RT	Product-standard value; recalled, standard text not re-read	Medium	Conditionally verified	H900 is higher strength/lower toughness; H1150 lower strength/higher toughness.
P17	41	SS 430 (A240 Type 430)	Annealed plate/sheet (specified)	Specified minimum	7,740	7.740	200.0	0.280	78.1	205	450		22	≤183 HBW (A240 max)	26.1	10.4	6.00E-07	UTS	58.1	25.84		RT	Product-standard value; recalled, standard text not re-read	Medium	Conditionally verified	
P18	42	SS 440C	Hardened & tempered ≈150–315 °C	Typical	7,650	7.650	200.0							≈57–60 HRC				—		26.14		RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	Hardness-only record.
P19	47	AISI D2	Hardened & tempered (working hardness)	Typical	7,700	7.700								≈58–62 HRC				—				RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	Hardness depends on tempering temperature.
P20	49	AISI H13	Hardened & tempered (hot-work practice)	Typical	7,800	7.800								≈44–52 HRC				—				RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	
P21	50	AISI M2	Hardened & tempered	Typical	8,150	8.150								≈62–65 HRC				—				RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	
P22	45	AISI O1	Hardened & tempered	Typical	7,810	7.810								≈57–62 HRC				—				RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	
P23	46	AISI A2	Hardened & tempered	Typical	7,860	7.860								≈57–62 HRC				—				RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	
P24	56	IS 210 FG 150	Separately cast test bar (specified)	Specified minimum	7,100	7.100	85.0	0.260	33.7		150			casting-section dependent				UTS	21.1	11.97	No distinct yield; brittle (<1% elong.); compressive strength ≈3–4× tensile	RT	Product-standard value; recalled, standard text not re-read	Low	Unverified	p, E approximate class-range midpoints; E depends on stress level.
P25	57	IS 210 FG 200	Separately cast test bar (specified)	Specified minimum	7,100	7.100	100.0	0.260	39.7		200			≈150–220 HBW (typical)				UTS	28.2	14.08	Damping high vs steel	RT	Product-standard value; recalled, standard text not re-read	Low	Unverified	As FG 150.
P26	58	IS 210 FG 260	Separately cast test bar (specified)	Specified minimum	7,100	7.100	115.0	0.260	45.6		260			casting-section dependent				UTS	36.6	16.20		RT	Product-standard value; recalled, standard text not re-read	Low	Unverified	As FG 150.
P27	59	EN-GJL-200	Separately cast bar (specified min)	Specified minimum	7,100	7.100	100.0	0.260	39.7		200			informative only				UTS	28.2	14.08		RT	Product-standard value; recalled, standard text not re-read	Low	Unverified	EN 1561 gives Rm range; E approximate.
P28	60	EN-GJL-250	Separately cast bar (specified min)	Specified minimum	7,100	7.100	115.0	0.260	45.6		250			informative only				UTS	35.2	16.20		RT	Product-standard value; recalled, standard text not re-read	Low	Unverified	
P29	64	ASTM A48 Class 30	Separately cast bar (specified)	Specified minimum	7,100	7.100	100.0	0.260	39.7		207			n/a				UTS	29.2	14.08	30 ksi min tensile	RT	Product-standard value; recalled, standard text not re-read	Low	Unverified	E approximate class-range midpoint.
P30	61	EN-GJS-400-15	Separately cast bar (specified min)	Specified minimum	7,100	7.100	169.0	0.275	66.3	250	400		15	≈135–180 HBW (informative)				UTS	56.3	23.80		RT	Product-standard value; recalled, standard text not re-read	Medium	Conditionally verified	Rp0.2 / Rm / A minima.
P31	62	EN-GJS-500-7	Separately cast bar (specified min)	Specified minimum	7,100	7.100	169.0	0.275	66.3	320	500		7	≈170–230 HBW (informative)				UTS	70.4	23.80		RT	Product-standard value; recalled, standard text not re-read	Medium	Conditionally verified	
P32	63	ASTM A536 65-45-12	Specified	Specified minimum	7,100	7.100	169.0	0.275	66.3	310	448		12	typ. ≈156–217 HBW				UTS	63.1	23.80	65/45 ksi	RT	Product-standard value; recalled, standard text not re-read	Medium	Conditionally verified	
P33	72	AA 6061-T6	Sheet/plate/extrusion (typical)	Typical	2,700	2.700	68.9	0.330	25.9	276	310		12	≈95 HBW	167	23.6	3.99E-08	UTS	114.8	25.52	Fatigue ≈96.5 MPa at 5×10^8 cycles (R. R. Moore, R=−1)	RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	ASTM minima are lower (e.g. ≈240/260 MPa for some extrusions).
P34	76	AA 7075-T6	Sheet/plate (typical)	Typical	2,810	2.810	71.7	0.330	27.0	503	572		11	≈150 HBW	130	23.4	5.15E-08	UTS	203.6	25.52	Fatigue ≈159 MPa at 5×10^8 cycles; KIC ≈29 MPa√m (T651, L-T, approx.)	RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	SCC-susceptible short-transverse.
P35	68	AA 2024-T3	Sheet (typical)	Typical	2,780	2.780	73.1	0.330	27.5	345	483		18	≈120 HBW	121	23.2	5.82E-08	UTS	173.7	26.29	Fatigue ≈138 MPa at 5×10^8 cycles	RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	
P36	67	AA 2014-T6	Bar/forging (typical)	Typical	2,800	2.800	72.4	0.330	27.2	414	483		13	≈135 HBW	154	23		UTS	172.5	25.86		RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	
P37	70	AA 5052-H32	Sheet (typical)	Typical	2,680	2.680	70.3	0.330	26.4	193	228		12	≈60 HBW	138	23.8	4.95E-08	UTS	85.1	26.23	Fatigue ≈117 MPa at 5×10^8 cycles	RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	
P38	71	AA 5083-H111	EN 485-2 minima	Specified minimum	2,660	2.660	71.0	0.330	26.7	125	275		15	≈75 HBW (approx.)	117	23.8	5.90E-08	UTS	103.4	26.69	UTS range 275–350 MPa	RT	Product-standard value; recalled, standard text not re-read	Medium	Conditionally verified	ASTM tempers (H116/H321) have higher yield.
P39	69	AA 3003-H14	Sheet (typical)	Typical	2,730	2.730	68.9	0.330	25.9	145	150		8	≈40 HBW	159	23.2		UTS	54.9	25.24		RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	
P40	74	AA 6063-T6	Extrusion (typical)	Typical	2,690	2.690	68.9	0.330	25.9	214	241		12	≈73 HBW	200	23.4		UTS	89.6	25.61		RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	
P41	77	A356.0-T6	Sand cast (typical)	Typical	2,680	2.680	72.4	0.330	27.2	186	262		5	≈80 HBW	151	21.5		UTS	97.8	27.01	Fatigue/elongation highly porosity-dependent	RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	Casting vs test-bar properties differ.
P42	78	A380.0	Die cast (typical)	Typical	2,710	2.710	71.0	0.330	26.7	160	324		3.5	≈80 HBW	96	21.8		UTS	119.6	26.20		RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	
P43	80	UNS C11000 (ETP Cu)	Annealed O60	Typical	8,940	8.940	117.0	0.340	43.7	69	220		45	n/a	391	17	1.72E-08	UTS	24.6	13.09	≥100% IACS (1.7241×10^-8 Ω·m at 20 °C)	RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	High	Conditionally verified	
P44	81	UNS C26000	Annealed O61	Typical	8,530	8.530	110.0			105	325		62	n/a	120	19.9	6.20E-08	UTS	38.1	12.90	≈28% IACS	RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	
P45	84	UNS C93200	Sand cast (B584 min)	Specified minimum	7,600	7.600	103.0			124	241		15	≈65 HBW (500 kg)	59	18		UTS	31.7	13.55		RT	Product-standard value; recalled, standard text not re-read	Medium	Conditionally verified	
P46	85	UNS C95400	As-cast (B148 min)	Specified minimum	7,450	7.450	110.0			207	517		12	≈150 HBW (3000 kg, typical)	59	16.2		UTS	69.4	14.77		RT	Product-standard value; recalled, standard text not re-read	Medium	Conditionally verified	
P47	87	Ti Grade 2	Plate/sheet (B265 min)	Specified minimum	4,510	4.510	105.0	0.370	38.3	275	345		20	n/a	16.4	8.6	5.20E-07	UTS	76.5	23.28		RT	Product-standard value; recalled, standard text not re-read	Medium	Conditionally verified	Max yield 450 MPa.
P48	88	Ti-6Al-4V Gr 5	Annealed (B265 min)	Specified minimum	4,430	4.430	113.8	0.342	42.4	825	895		10	≈36 HRC (typical)	6.7	8.6	1.70E-06	UTS	202.0	25.69	KIC ≈55–75 MPa√m depending on microstructure	RT	Handbook typical (ASM/MatWeb-type); recalled, standard text not re-read	Medium	Conditionally verified	Typical YS/UTS ≈880/950 MPa.
P49	89	AZ31B	O-temper sheet (typical)	Typical	1,770	1.770	45.0	0.350	16.7	150	255		21	n/a	96	26	9.20E-08	UTS	144.1	25.42		RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	
P50	90	AZ91D	Die cast (typical)	Typical	1,810	1.810	45.0	0.350	16.7	160	230		3	n/a	72	25		UTS	127.1	24.86		RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	
P51	91	UNS N06600	Annealed (B168 min)	Specified minimum	8,470	8.470	207.0			240	550		30	n/a	14.9	13.3	1.03E-06	UTS	64.9	24.44		RT	Product-standard value; recalled, standard text not re-read	Medium	Conditionally verified	
P52	92	UNS N06625	Annealed Gr 1 (B443 min)	Specified minimum	8,440	8.440	205.0			414	827		30	n/a	9.8	12.8	1.29E-06	UTS	98.0	24.29		RT	Product-standard value; recalled, standard text not re-read	Medium	Conditionally verified	
P53	93	UNS N07718	Solution + aged (AMS-type minimum, approx.)	Specified minimum	8,190	8.190	200.0	0.290	77.5	1034	1275		12	n/a	11.4	13	1.25E-06	UTS	155.7	24.42	Creep/rupture data required for >650 °C	RT	Product-standard value; recalled, standard text not re-read	Low	Unverified	Exact minima depend on specification and form.
P54	94	UNS N04400	Annealed (B164/B127 min)	Specified minimum	8,800	8.800	179.0			195	480		35	n/a	21.8	13.9	5.10E-07	UTS	54.5	20.34		RT	Product-standard value; recalled, standard text not re-read	Medium	Conditionally verified	
P55	95	UNS N10276	Solution annealed (B575 min)	Specified minimum	8,890	8.890	205.0			283	690		40	n/a	10.2	11.2	1.30E-06	UTS	77.6	23.06		RT	Product-standard value; recalled, standard text not re-read	Medium	Conditionally verified	
P56	96	Zamak 3 (AG40A)	Die cast (typical)	Typical	6,600	6.600	85.0	0.300	32.7	221	283		10	≈82 HBW	113	27.4		UTS	42.9	12.88	Creep at RT is significant	RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	
P57	97	Zamak 5 (AC41A)	Die cast (typical)	Typical	6,600	6.600	85.0	0.300	32.7	269	331		7	≈91 HBW	109	27.4		UTS	50.2	12.88		RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	
P58	98	PA6	Dry as moulded, unfilled	Typical (range)	1,140	1.140	2.8				75			n/a	0.25	80		UTS	65.8	2.46	Properties fall markedly with moisture uptake	23 °C	Manufacturer datasheet typical; recalled	Low	Unverified	Grade/moisture dependent.
P59	99	PA66	Dry as moulded, unfilled	Typical (range)	1,140	1.140	3.0				80			n/a	0.25	80		UTS	70.2	2.63	As PA6	23 °C	Manufacturer datasheet typical; recalled	Low	Unverified	
P60	100	POM-C	Unfilled	Typical (range)	1,410	1.410	2.8			62			40	n/a	0.31	110		YS	44.0	1.99		23 °C	Manufacturer datasheet typical; recalled	Low	Unverified	
P61	101	POM-H	Unfilled	Typical (range)	1,420	1.420	3.1			70			35	n/a		110		YS	49.3	2.18		23 °C	Manufacturer datasheet typical; recalled	Low	Unverified	
P62	102	PTFE	Virgin	Typical (range)	2,150	2.150	0.5				25		300	n/a	0.25	120		UTS	11.6	0.23	Cold flow (creep) significant	23 °C	Manufacturer datasheet typical; recalled	Low	Unverified	
P63	103	PEEK	Unfilled	Typical (range)	1,300	1.300	3.6			100			30	n/a	0.25	47		YS	76.9	2.77	Tg ≈143 °C; Tm ≈343 °C	23 °C	Manufacturer datasheet typical; recalled	Low	Unverified	
P64	104	ABS	General purpose	Typical (range)	1,050	1.050	2.2			40				n/a	0.17	90		YS	38.1	2.10		23 °C	Manufacturer datasheet typical; recalled	Low	Unverified	
P65	105	Polycarbonate (PC)	Unfilled	Typical (range)	1,200	1.200	2.3			62			100	n/a	0.2	68		YS	51.7	1.92		23 °C	Manufacturer datasheet typical; recalled	Low	Unverified	

Prop ID	Master No.	Designation	Condition / product form	Value basis	Density ρ (kg/m³)	Density (g/cm³) [calc]	Young's modulus E (GPa)	Poisson ν	Shear modulus G (GPa) [derived,	Yield / proof strength (MPa)	Ultimate tensile strength (MPa)	Flexural strength (MPa)	Elongation at fracture (%)	Hardness (as stated)	Thermal conductivity k (W/(m·K))	CTE (µm/(m·K))	Electrical resistivity (Ω·m)	Strength used [calc]	Specific strength (kN·m/kg) [calc]	Specific stiffness (MN·m/kg) [calc]	Impact / fatigue / KIC / other	Test temperature	Source class	Confidence	Verification status	Limitations / notes
P66	106	UHMWPE	Moulded/extruded	Typical (range)	935	0.935	0.7			21			350	n/a	0.4			YS	22.5	0.75		23 °C	Manufacturer datasheet typical; recalled	Low	Unverified	
P67	112	Alumina (99.5%)	Sintered	Typical (range)	3,890	3.890	370.0	0.220	151.6			350		=1500 HV	30	8		Flexural	90.0	95.12	KIC ≈4 MPa√m; Weibull statistics govern strength	RT	Manufacturer datasheet typical; recalled	Low	Unverified	Purity-dependent.
P68	113	SiC (sintered)	Sintered α-SiC	Typical (range)	3,150	3.150	410.0	0.170	175.2			400		≈2500 HV	120	4		Flexural	127.0	130.16	KIC ≈4 MPa√m	RT	Manufacturer datasheet typical; recalled	Low	Unverified	Route-dependent.
P69	114	ZrO2 (Y-TZP)	Sintered	Typical (range)	6,050	6.050	205.0	0.300	78.8			1000		≈1250 HV	2.5	10.5		Flexural	165.3	33.88	KIC ≈6–10 MPa√m (method dependent)	RT	Manufacturer datasheet typical; recalled	Low	Unverified	
P70	115	Si3N4 (gas-pressure sintered)	Sintered	Typical (range)	3,200	3.200	310.0	0.270	122.0			850		≈1500 HV	28	3.2		Flexural	265.6	96.88	KIC ≈6 MPa√m	RT	Manufacturer datasheet typical; recalled	Low	Unverified	
P71	116	WC-Co (≈6% Co)	Sintered	Typical (range)	14,900	14.900	620.0	0.220	254.1			2000		≈1500 HV	100	5.5		Flexural	134.2	41.61	TRS (flexural) shown; KIC ≈10–12 MPa√m	RT	Manufacturer datasheet typical; recalled	Low	Unverified	
P72	119	CFRP UD epoxy (Vf=0.6)	0° direction	Typical (range)	1,560	1.560	135.0				1800		1.4	n/a		-0.3		UTS	1153.8	86.54	E2 ≈9 GPa, G12 ≈5 GPa, ν12 ≈0.3; compression ≈1200–1500 MPa; ILSS ≈70–90 MPa	RT	Manufacturer datasheet typical; recalled	Low	Unverified	Strongly anisotropic — values only along fibre.
P73	120	GFRP UD E-glass/epoxy (Vf≈0.55)	0° direction	Typical (range)	1,950	1.950	40.0				900		2.5	n/a				UTS	461.5	20.51	Transverse values far lower	RT	Manufacturer datasheet typical; recalled	Low	Unverified	Strongly anisotropic.
P74	125	Invar 36	Annealed	Typical	8,050	8.050	141.0	0.260	56.0					n/a	10.5	1.3	7.50E-07	—		17.52	CTE rises above Curie temperature (≈230 °C)	RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	CTE value applies near RT.
P75	126	Kovar	Annealed	Typical	8,360	8.360	138.0							n/a	17	5.5	4.90E-07	—		16.51	CTE 30–400 °C approx.	RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	
P76	127	Molybdenum	Wrought	Typical	10,220	10.220	329.0	0.310	125.6					n/a	138	4.8	5.34E-08	—		32.19	Oxidises rapidly in air above ≈500–600 °C	RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	
P77	128	Tungsten	Wrought/sintered	Typical	19,250	19.250	411.0	0.280	160.5					n/a	173	4.5	5.28E-08	—		21.35		RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Medium	Conditionally verified	
P78	130	Austenitic manganese steel (A128)	Water-toughened	Typical	7,800	7.800								≈200 HBW; work-hardened surface ≈400–550 HBW				—			Hardening requires impact/deformation	RT	Handbook typical (ASM/MatWeb-type); recalled, not re-fetched	Low	Unverified	Grade-dependent.
P79	131	Hardox 400	Plate (manufacturer typical)	Manufacturer typical	7,850	7.850	200.0	0.300	76.9	1000	1250		10	370–430 HBW (SSAB)				UTS	159.2	25.48		RT	Manufacturer datasheet typical; recalled	Medium	Conditionally verified	E, ν generic steel; check current datasheet.
P80	132	Hardox 450	Plate (manufacturer typical)	Manufacturer typical	7,850	7.850	200.0	0.300	76.9	1200	1400		10	425–475 HBW (SSAB)				UTS	178.3	25.48		RT	Manufacturer datasheet typical; recalled	Low	Unverified	Check current datasheet.

Sheet 4 — Grain, Crystal & Microstructure Reference (family level)

Numbers in [ ] are Master No. from the upload. Grain size and phases depend on processing; no family-wide single value is implied. Statements are textbook-level knowledge, not re-checked against sources.

ID	Family / group	Representative grades [Master No.]	Crystal structure at RT	Phases / constituents	Typical grain / microstructure	Strengthening mechanisms	Heat-treatment / processing response	Structure → property effects & key risks	Metallography / grain-size method	Verification status
GS01	Low-carbon / structural steels	1006–1025, A36, IS 2062 E250 [1–6,12,14]	BCC α-ferrite (+ orthorhombic Fe3C)	Ferrite + pearlite	Equiaxed ferrite (hot-rolled/normalised); elongated + dislocation-dense (cold-drawn)	Grain-boundary (Hall–Petch), solid solution (Mn, Si), work hardening	Normalise to refine grain; cold work raises strength; limited hardenability	Fine grain → higher yield and lower ductile-brittle transition temperature (BCC DBTT); banding and decarburisation are quality risks	ASTM E112 grain size; E3/E407 prep/etch; E45 inclusions	Conditionally verified
GS02	Medium/high-carbon steels	1035–1060, 65Mn [7–11,55]	BCC α-ferrite + Fe3C; BCT martensite when quenched	Pearlite fraction rises with C	Pearlite colonies/lamellar spacing; spheroidised for machinability	Pearlite spacing, martensite (carbon), tempering carbides	Q&T; induction hardening; as-quenched BCT c/a =1+0.045·%C (approx.)	Fine pearlite = strength; coarse = ductility; quench cracking above ≈0.4–0.5%C in water	E112; hardness traverse; E3	Conditionally verified
GS03	HSLA structural steels	E350, A572 Gr50 [13,15]	BCC ferrite + pearlite/bainite	Nb/V/Ti carbonitrides	Very fine ferrite from controlled rolling (TMCP)	Grain refinement, precipitation, solid solution	Controlled rolling/accelerated cooling; no Q&T	HAZ grain coarsening/ MA constituents reduce toughness; weld procedure & CE control	E112; Charpy per grade	Conditionally verified
GS04	Q&T Cr-Mo / Ni-Cr-Mo steels	4130, 4140, 4340, 42CrMo4, 34CrNiMo6, 25CrMo4 [16–18,25,26]	BCT martensite (as-quenched) → tempered martensite (BCC + carbides)	Retained austenite (FCC) minor; bainite in thick sections	Lath martensite packets within prior-austenite grains	Martensite carbon, carbide precipitation, dislocation density	Austenitise → quench → temper; temper temp sets strength/toughness	Temper embrittlement (≈375–575 °C) and tempered-martensite embrittlement (≈250–400 °C); hydrogen embrittlement rises with strength; prior-austenite grain size controls toughness	E112 (PAGS); E3; hardness; Charpy	Conditionally verified
GS05	Case-hardening steels	8620, 9310, 16MnCr5, 20MnCr5, 18CrNiMo7-6 [19–22,27]	Case: BCT/BCC tempered martensite + carbides + retained austenite (FCC); core: low-C martensite/bainite	Carbide networks, retained austenite, intergranular oxidation in case	Case/core gradient (case depth typically ≈0.5–2 mm for gears)	Carbon gradient, martensite, compressive residual stress	Carburise ≈880–950 °C → quench → low temper; or carbonitride/nitride	Contact/bending fatigue depend on case depth, retained austenite, inclusions; grinding burn	ISO 2639 / E1077 case/decarb; microhardness E384; E3	Conditionally verified
GS06	Bearing steel	52100 / 100Cr6 [52]	Tempered martensite + spheroidal (Fe,Cr)3C carbides	Retained austenite (few %), carbides ≈ 0.5–1 µm	Fine, uniform carbide dispersion	Martensite + carbides	Spheroidise anneal → harden → temper (≈60–66 HRC)	Rolling-contact fatigue controlled by inclusions; white-etching cracks	E45/ISO 4967 cleanliness; E3; hardness	Conditionally verified
GS07	Spring steels	5160, 9254, 65Mn [53–55]	Tempered martensite (BCC + carbides)	Minor retained austenite	Fine tempered martensite; surface must be free of decarb	Martensite + tempering	Oil quench + temper (≈42–52 HRC typical)	Fatigue initiates at surface defects/decarburisation; shot peening helps	E1077 decarb; hardness	Conditionally verified
GS08	Tool steels (cold-work)	O1, A2, D2 [45–47]	BCT martensite + carbides (cementite / M7C3)	Retained austenite (FCC)	Carbide size/banding strongly alloy-dependent (D2 coarse)	Martensite + hard carbides	Harden + multiple tempers; D2 secondary hardening	Chipping (D2), distortion, quench cracking	E3; hardness; carbide rating	Conditionally verified
GS09	Tool steels (hot-work & HSS)	H11, H13, M2, M35 [48–51]	Tempered martensite + MC/M6C/M2C carbides	Retained austenite converted by multiple tempers	Fine carbides, secondary hardening	Secondary hardening carbides	Austenitise → quench → 2–3 tempers	Thermal fatigue (heat checking), temper softening, banding	E3; hardness; microstructure rating	Conditionally verified
GS10	Austenitic stainless	201, 301, 304/L, 316/L, 321, 310S [31–38]	FCC γ-austenite	Possible δ-ferrite, strain-induced α′ martensite, carbides, sigma (long exposure)	Equiaxed twinned grains	Solid solution (N, Mo, C), cold work	Solution anneal & quench; cold work for strength (301)	No DBTT (FCC); sensitisation ≈450–850 °C; chloride SCC; strain-induced martensite makes 301/304 slightly magnetic after cold work	ASTM A262 IGC; E112; ferrite number	Conditionally verified
GS11	Ferritic / martensitic stainless	410, 420, 430, 440C [39–42]	BCC ferrite (430); BCT martensite (410/420/440C)	Cr carbides; 440C has large primary carbides	Coarse-grain risk in ferritic after welding	Martensite (C), carbides	Harden + temper (martensitic); anneal (ferritic)	Reduced corrosion resistance if tempered in sensitising range; ferritic DBTT near RT in thick sections	E3; hardness; salt-spray/ASTM A262-type tests as applicable	Conditionally verified
GS12	Duplex & PH stainless	2205, 17-4 PH [43,44]	Duplex: BCC δ + FCC γ (~50:50). 17-4: martensite (BCC/BCT) + Cu-rich precipitates	Sigma/chi phases, α′ (475 °C), carbides; delta ferrite stringers in 17-4	Fine duplex lamellar grains aligned in rolling direction	Grain refinement + N (duplex); precipitation (17-4)	Solution anneal; ageing H900–H1150 for 17-4	Sigma/475 °C embrittlement; duplex low-temperature toughness depends on phase balance; 17-4 SCC/hydrogen risk in H900	Ferrite measurement (ASTM E562/E1181-type practice, verify); A923 (verify)	Conditionally verified
GS13	Gray cast iron	IS 210 FG 150/200/260, EN-GJL, A48 [56–60,64]	Graphite flakes (hex) in ferrite/pearlite (BCC) matrix	Steadite (Fe–Fe3P) possible; cementite (chill)	Flake type/size (ASTM A247 types A–E)	Matrix pearlite, flake size	Stress-relief; alloying (Cu, Cr, Mo) raises strength	Flakes = internal notches: low tensile, high damping, excellent compressive strength; section-size sensitivity	ASTM A247 graphite; E3; Brinell	Conditionally verified
GS14	Ductile (SG) iron	EN-GJS-400-15, 500-7, A536 [61–63]	Graphite nodules in ferrite and/or pearlite (BCC) matrix	Carbides, dross, exploded graphite defects	Nodule count/size; ferrite 'bull's-eye'	Matrix + nodularity	Mg/Ce treatment; ferritise anneal; ADI austemper	Nodularity ≥ typical 80–90% for rated grades; slag/shrinkage defects reduce fatigue and toughness	ASTM A247 nodularity; E3; tensile	Conditionally verified
GS15	Aluminium non-heat-treatable	1050, 1100, 3003, 5052, 5083 [65,66,69–71]	FCC α-Al	Dispersoids (Mn, Fe), Mg in solution	Equiaxed (O temper) or elongated (H tempers)	Cold work, solid solution (Mg)	Strain hardening; anneal	No DBTT; 5xxx sensitisation (>=65 °C long-term) → IGC/SCC; welds soften to O-like strength in work-hardened tempers	E112; ASTM G67 NAMLT (verify)	Conditionally verified
GS16	Aluminium heat-treatable	2014, 2024, 6061, 6063, 6082, 7075 [67,68,72–76]	FCC α-Al matrix + precipitates	θ′(Al2Cu), β″/β′(Mg2Si), η′/η(MgZn2), PFZ, grain-boundary precipitates	Recrystallised vs unrecrystallised/pancaked grains; anisotropy	Precipitation hardening	Solution treat → quench → age (T6) or overage (T73) for SCC	7xxx short-transverse SCC and exfoliation; 6061-T6 weld HAZ softening; quench sensitivity/distortion	Conductivity & hardness for temper check; ASTM G34/G47 (verify)	Conditionally verified
GS17	Aluminium cast alloys	A356, A380 [77,78]	FCC α-Al dendrites + eutectic Si (diamond cubic)	Fe-rich intermetallics, Mg2Si, porosity	Dendrite arm spacing; modified (Sr) fibrous Si	Precipitation (A356-T6), eutectic Si	Sr modification, degas, T6; HIP for critical parts	Fatigue controlled by pore/oxide size; die-cast skin/core	E3; porosity: radiography/CT	Conditionally verified
GS18	Copper & brasses	C10100, C11000, C26000, C36000 [79–82]	FCC Cu / FCC α-brass; α + BCC β′ in C36000	Pb globules (C36000), Cu2O (C11000)	Twinned equiaxed grains	Cold work, solid solution (Zn)	Anneal/cold-work tempers	Conductivity falls with impurities/cold work; dezincification, season cracking	ASTM E112 (copper grain size E112), conductivity (IACS)	Conditionally verified
GS19	Bronzes	C51000, C93200, C95400, C86300 [83–86]	FCC α (+ δ, κ, β depending on alloy)	Pb islands (C93200), κ phases (C95400), β (C86300)	Cast dendritic or wrought equiaxed	Solid solution, intermetallics, cold work	Heat-treatable for some Al-bronzes; cast alloys as-cast	Bearing bronze wear relies on soft Pb + hard δ; Al-bronze dealuminification	E3; hardness	Conditionally verified
GS20	Titanium alloys	Gr 2, Ti-6Al-4V [87,88]	HCP α (Gr 2); α (HCP) + β (BCC) in Ti-6Al-4V	α-case, TiH, ω/α₂ in other alloys	Equiaxed α, lamellar, or bimodal; prior-β grain size	Solid solution (O, Al), α/β phase fraction & morphology	Mill anneal, STA, β-process	Oxygen/hydrogen embrittlement; alpha-case; microstructure sets fracture-toughness/fatigue trade-off; β transus ≈995 °C	E112 not applicable to lamellar; ASTM E2333/F1108-type practices (verify); E3	Conditionally verified
GS21	Magnesium alloys	AZ31B, AZ91D [89,90]	HCP α-Mg (+ β-Mg17Al12)	Mg17Al12 at grain boundaries, Al-Mn intermetallics	Wrought: recrystallised with basal texture; die-cast: fine skin + porous core	Solid solution (Al, Zn), grain refinement, precipitates	Limited heat treatment; rolled/extruded anneals	Low RT formability (few slip systems), anisotropy; galvanic corrosion; flammable fines	E112; E3	Conditionally verified
GS22	Nickel alloys	600, 625, 718, Monel 400, C-276 [91–95]	FCC γ (+ γ″ BCT, γ′ L1_2, δ orthorhombic in 718)	Carbides, Laves, δ; σ/μ/P on long exposure	Twinned equiaxed grains; grain size critical for creep and notch	Solid solution (Mo, Nb, Cr), precipitation (718)	Solution anneal; double age (718)	No DBTT; creep strength from γ″/γ′ and grain-boundary carbides; 718 loses strength >=650 °C	E112; E3; creep E139	Conditionally verified
GS23	Zinc die-cast alloys	Zamak 3, 5 [96,97]	HCP η-Zn + Al-rich FCC phase	Zn-Al eutectoid decomposition (ageing)	Fine dendritic die-cast structure	Solid solution, fine grain	None (as-cast)	Room-temperature creep, dimensional ageing, Pb/Cd/Sn impurity-driven intergranular corrosion	E3	Conditionally verified
GS24	Engineering polymers	PA6, PA66, POM, PTFE, PEEK, UHMWPE, HDPE, PET, PPS [98–103,106,107,110,111]	Semi-crystalline: crystalline lamellae + amorphous regions	Spherulites, tie molecules, fillers	Crystallinity 30–80% typical depending on polymer & cooling	Crystallinity, molecular weight, orientation, fillers	Moulding/annealing controls crystallinity	Moisture (PA), creep, Tg/Tm limits, notch sensitivity, UV/chemical attack	DSC (E1356/E794 etc.), ASTM D638, D256	Conditionally verified
GS25	Amorphous polymers	ABS, PC, PVC-U [104,105,108]	Amorphous (no long-range order)	Rubber phase in ABS	Glassy below Tg; flow orientation	Chain entanglement, rubber toughening	Moulding/annealing	Environmental stress cracking, notch brittleness, weld-line weakness	D638; D648 HDT; D256	Conditionally verified
GS26	Elastomers (TPU)	TPU [109]	Phase-separated hard/soft block copolymer	Hard-domain crystallinity	Micro-phase separated	Physical crosslinks	Processing temp / annealing	Hysteresis/heat build-up; hydrolysis (polyester TPU)	D412, D2240	Conditionally verified
GS27	Technical ceramics	Al2O3, SiC, ZrO2, Si3N4 [112–115]	Alumina: hexagonal/trigonal corundum; SiC: hexagonal α/cubic β; ZrO2: monoclinic/tetragonal/cubic; Si3N4: hexagonal β/α	Glassy grain-boundary films, porosity	Grain size = sub-µm to tens of µm; elongated β-Si3N4	Grain size, toughening mechanisms (transformation toughening in Y-TZP)	Sintering (pressureless, GPS, HIP)	Brittle, flaw-controlled strength (Weibull), thermal shock, low-temperature ageing in Y-TZP	ASTM C1161, C1421, C1327	Conditionally verified
GS28	Hardmetals & superabrasives	WC-Co, CBN, PCD [116–118]	WC hexagonal + Co binder (FCC/HCP); cBN cubic; diamond cubic	Eta phase (brittle) if carbon-deficient; binder pools	WC grain size = 0.2–6 µm typical	Grain size, binder fraction	Liquid-phase sintering/HPHT	Hardness–toughness trade-off; thermal/chemical wear (PCD on ferrous)	ASTM B276 microstructure (verify), B311, B406	Conditionally verified
GS29	Composites	CFRP, GFRP, aramid [119–121]	Fibre: carbon (turbostratic graphitic), glass (amorphous), aramid (oriented polymer); matrix polymer	Voids, fibre waviness, resin-rich zones	Layup-dependent laminate microstructure; fibre Vf = 50–65% typical	Fibre-dominated; matrix controls shear/compression/Tg	Cure cycle controls void content and Tg	Delamination, impact damage (BVID), moisture, galvanic corrosion with metals (carbon)	ASTM D3171, D2734 (verify), D3039, D7136 (verify)	Conditionally verified
GS30	Shape-memory & special alloys	NiTi, CuBe, CoCr, Invar, Kovar [122–126]	NiTi: B2 ↔ B19′; CuBe: FCC + precipitates; CoCr: FCC/HCP; Invar/Kovar: FCC	Ni4Ti3 precipitates (NiTi), γ′ (CuBe), carbides (CoCr)	Fine recrystallised grains	Precipitation / transformation	Solution + age; cold work + anneal (NiTi); stabilising anneal (Invar)	Af temperature sensitive to ±0.1 at% Ni; Be hazard; Invar creep/dimension drift	DSC (F2004), dilatometry (E228)	Conditionally verified
GS31	Refractory metals & graphite	Mo, W, graphite [127–129]	BCC (W, Mo); hexagonal layered graphite	Oxides; porosity in P/M	Elongated worked grains; recrystallised after high T	Work hardening, grain size	Powder metallurgy + working	W/Mo brittle near RT (high DBTT), oxidise in air; graphite oxidises >=400–500 °C	E112; density	Conditionally verified
GS32	Wear-resistant steels	Hadfield, Hardox 400/450 [130–132]	Hadfield: FCC austenite (+carbides); Hardox: martensite/bainite (BCC/BCT)	Carbide networks (Hadfield as-cast), retained austenite	Water-toughened austenite; fine Q&T lath structure	Strain-induced twinning/work hardening (Hadfield); martensite (Hardox)	Water toughen (Hadfield); Q or Q&T (Hardox)	Hadfield needs impact to harden; Hardox HAZ softening	Hardness; E3; Charpy	Conditionally verified
GS33	Additive-manufactured metals	AlSi10Mg, 316L, Ti64, IN718 (gap list)	Same crystal structures as wrought but non-equilibrium microstructure	Porosity, lack-of-fusion, α′ martensite (Ti64), cellular segregation	Columnar grains along build direction; very fine cells; strong texture	Cellular structure, grain refinement, dislocations	Stress relief, HIP, solution/age	Anisotropy (build direction), rough surface lowers fatigue, residual stress, powder reuse (O pickup)	ASTM E3/E112; F3122 (verify); CT/density	Conditionally verified

Crystal-structure fundamentals

#	Structure	Examples	Coordination / packing	Slip systems	Behaviour
1	BCC	α-Fe, Cr, Mo, W, V; ferritic steels	CN 8, APF 0.68	12 primary {110}<111> (up to 48 incl. {112},{123})	Yes — ductile-to-brittle transition; strength rises strongly at low T/high strain rate

ID	Family / group	Representative grades [Master No.]	Crystal structure at RT	Phases / constituents	Typical grain / microstructure	Strengthening mechanisms	Heat-treatment / processing response	Structure → property effects & key risks	Metallography / grain-size method	Verification status
2	FCC	γ-Fe, Al, Cu, Ni, Pb; austenitic SS, Ni alloys	CN 12, APF 0.74	12 {111}<110>	No DBTT — ductile at cryogenic T; high work-hardening					
3	HCP	Ti (α), Mg, Zn, Zr, Be, Co (ε)	CN 12, APF 0.74 (ideal c/a)	Basal <a> (3) at RT; prismatic/pyramidal depend on metal & T	Limited RT slip in Mg/Zn → anisotropy, twinning; Ti has more slip modes					
4	BCT (martensite)	Fast-quenched steel (C trapped in octahedral sites)	c/a = 1 + 0.045·wt%C (approx.)	Distorted BCC; very high dislocation density	Hard/brittle until tempered (→ BCC + carbides)					

Reference transformation temperatures / compositions (approximate)

#	Item	Approximate value
1	α-Fe ↔ γ-Fe transformation (pure Fe)	≈912 °C (BCC→FCC); δ-Fe BCC ≈1394–1538 °C
2	Fe–C eutectoid	≈727 °C, ≈0.76 wt% C
3	Fe–C eutectic	≈1147 °C, ≈4.3 wt% C
4	Max C solubility	≈0.022 wt% in ferrite (727 °C); ≈2.14 wt% in austenite (1147 °C)
5	Ti β transus	Pure Ti ≈882 °C; Ti-6Al-4V ≈995 °C (typical, composition dependent)
6	Al–Si eutectic	≈577 °C, ≈12.6 wt% Si
7	Sn (white tin)	β-Sn BCT at RT; α-tin (diamond cubic) below ≈13 °C
8	Co	HCP (ε) at RT; FCC above ≈417 °C

Sheet 5 — Formula Register & Worked Examples

Yellow/blue = assumed input (change to re-run); green = linked from Sheet 3; grey = formula. Each block ends with an independent-path check and a computed status.

Block	Quantity / step	Symbol	Value	Unit	Type	Equation / assumptions / note	Status
B1	**Density & unit conversion**					**Equation: ρ = m / V ; 1 g/cm³ = 1000 kg/m³ (ρ in kg/m³, m in kg, V in m³)**	
	Assumptions / applicability					Homogeneous solid, no voids; mass and dimensions measured at same temperature.	
B1	Mass	m	0.785	kg	Input	Assumed example (steel block)	
B1	Length	Lx	0.1	m	Input		
B1	Width	Ly	0.05	m	Input		
B1	Height	Lz	0.02	m	Input		
B1	Volume	V	0.0001	m³	Calc	V = Lx·Ly·Lz	
B1	Density	ρ	7850	kg/m³	Calc	ρ = m/V	
B1	Density (converted)	ρ	7.85	g/cm³	Calc	Conversion factor 1000	
B1	Check: grams ÷ cm³	ρ_chk	7.85	g/cm³	Check	Independent path in g and cm	
B1	Verification tolerance (relative)	tol	0.005	–	Input	Pass if |check − result| / result ≤ tol	
B1	Independent check: |chk − rhog| / rhog	Δrel	0.000%	–	Status	Relative difference between primary and independent route	Verified
B2	**Linear thermal expansion**					**Equation: ΔL = α·L0·ΔT (α in K⁻¹; ΔT in K)**	
	Assumptions / applicability					Constant α is acceptable for modest ΔT (≈ <100–200 K) away from phase changes; use dL/L tables for wide ranges, cryogenic service, Invar/polymers.	
B2	Original length	L0	1	m	Input	Assumed example	
B2	CTE of AA 6061-T6 (Sheet 3)	α	23.6	µm/(m·K)	Linked	Linked	
B2	Temperature change	ΔT	80	K	Input	Assumed	
B2	Length change	ΔL	0.001888	m	Calc	α converted to K⁻¹	
B2	Length change	ΔL	1.888	mm	Calc		
B2	Check: µm route	ΔL_chk	1.888	mm	Check	α[µm/m·K]×L0[m]×ΔT[K] = µm	
B2	Verification tolerance (relative)	tol	0.005	–	Input	Pass if |check − result| / result ≤ tol	
B2	Independent check: |chk − dLmm| / dLmm	Δrel	0.000%	–	Status	Relative difference between primary and independent route	Verified
B3	**Electrical resistance of a conductor**					**Equation: R = ρe·L / A (ρe = electrical resistivity, Ω·m — NOT mass density)**	
	Assumptions / applicability					Uniform round wire, DC, 20 °C; temperature coefficient not included.	
B3	Resistivity (annealed Cu, 100% IACS)	ρe	1.724E-08	Ω·m	Input	International Annealed Copper Standard value	
B3	Length	L	100	m	Input	Assumed	
B3	Wire diameter	d	0.002	m	Input	Assumed	
B3	Cross-section	A	3.1415926536E-06	m²	Calc		
B3	Resistance	R	0.54879807476947	Ω	Calc		
B3	Check: Ω·mm²/m route	R_chk	0.54879807476947	Ω	Check	ρ converted to Ω·mm²/m	
B3	Verification tolerance (relative)	tol	0.005	–	Input	Pass if |check − result| / result ≤ tol	
B3	Independent check: |chk − R| / R	Δrel	0.000%	–	Status	Relative difference between primary and independent route	Verified
B4	**Specific strength & specific stiffness (AA 6061-T6)**					**Equation: σ/ρ and E/ρ — here σ = UTS (state the strength definition used)**	
	Assumptions / applicability					Consistent SI units; only meaningful for comparison at equal load path (see Material Selection indices).	
B4	UTS (Sheet 3)	σ	310	MPa	Linked	Linked	
B4	Density (Sheet 3)	ρ	2,700	kg/m³	Linked	Linked	
B4	E (Sheet 3)	E	68.9	GPa	Linked	Linked	
B4	Specific strength	σ/ρ	114.814814814815	kN·m/kg	Calc	MPa ÷ (kg/m³) = 1000 N·m/kg	

Block	Quantity / step	Symbol	Value	Unit	Type	Equation / assumptions / note	Status
B4	Specific stiffness	E/ρ	25.5185185185185	MN·m/kg	Calc		
B4	Check: MPa ÷ (g/cm³)	(σ/ρ)_chk	114.814814814815	kN·m/kg	Check	Numerically equal	
B4	Check: GPa ÷ (g/cm³)	(E/ρ)_chk	25.5185185185185	MN·m/kg	Check		
B4	Verification tolerance (relative)	tol	0.005	–	Input	Pass if |check − result| / result ≤ tol	
B4	Independent check: |chk − ss| / ss	Δrel	0.000%	–	Status	Relative difference between primary and independent route	Verified
B4	Independent check: |chk2 − sf| / sf	Δrel	0.000%	–	Status	Relative difference between primary and independent route	Verified
B5	**Engineering stress, strain, Hooke's law**					**Equation: σ = F/A0 ; ε = σ/E ; ΔL = ε·L0**	
	Assumptions / applicability					Axial, uniform stress, linear-elastic (σ below yield), small strain.	
B5	Axial force	F	20	kN	Input	Assumed	
B5	Original area	A0	100	mm²	Input	Assumed	
B5	Gauge length	L0	50	mm	Input	Assumed	
B5	E, AISI 1045 HR (Sheet 3)	E	205.0	GPa	Linked	Linked	
B5	Stress	σ	200	MPa	Calc	N/mm²	
B5	Strain	ε	0.0009756097561	–	Calc		
B5	Elongation	ΔL	0.04878048780488	mm	Calc		
B5	Check: SI route ΔL = F·L0/(A·E)	ΔL_chk	0.04878048780488	mm	Check	All SI then converted	
B5	Verification tolerance (relative)	tol	0.005	–	Input	Pass if |check − result| / result ≤ tol	
B5	Independent check: |chk − dL| / dL	Δrel	0.000%	–	Status	Relative difference between primary and independent route	Verified
B6	**Bending stress (rectangular beam)**					**Equation: σ = M·c / I ; I = b·h³/12 ; c = h/2**	
	Assumptions / applicability					Prismatic, linear-elastic, Euler–Bernoulli, no stress concentration, load in a principal plane.	
B6	Bending moment	M	1500	N·m	Input	Assumed	
B6	Width	b	0.03	m	Input		
B6	Depth	h	0.05	m	Input		
B6	Second moment of area	I	0.0000003125	m⁴	Calc		
B6	Distance to outer fibre	c	0.025	m	Calc		
B6	Max bending stress	σ	120	MPa	Calc		
B6	Check: σ = 6M/(b·h²)	σ_chk	120	MPa	Check	Section-modulus form	
B6	Verification tolerance (relative)	tol	0.005	–	Input	Pass if |check − result| / result ≤ tol	
B6	Independent check: |chk − sig| / sig	Δrel	0.000%	–	Status	Relative difference between primary and independent route	Verified
B7	**Torsional shear stress (solid round shaft)**					**Equation: τ = T·r / J ; J = π·d⁴/32**	
	Assumptions / applicability					Solid circular section, elastic, no keyway/shoulder stress concentration.	
B7	Torque	T	500	N·m	Input	Assumed	
B7	Diameter	d	0.03	m	Input		
B7	Polar moment	J	7.9521564044E-08	m⁴	Calc		
B7	Max shear stress	τ	94.3140403507528	MPa	Calc		
B7	Check: τ = 16T/(π·d³)	τ_chk	94.3140403507528	MPa	Check		
B7	Verification tolerance (relative)	tol	0.005	–	Input	Pass if |check − result| / result ≤ tol	
B7	Independent check: |chk − tau| / tau	Δrel	0.000%	–	Status	Relative difference between primary and independent route	Verified
B8	**Factor of safety (yield basis)**					**Equation: n = S_y / σ_working (ductile metals; use UTS or fracture basis for brittle materials)**	
	Assumptions / applicability					Static load; σ_working must include stress concentrations/dynamic factors for real design; compare with code-required factors.	

Block	Quantity / step	Symbol	Value	Unit	Type	Equation / assumptions / note	Status
B8	Yield, AISI 4140 Q&T (Sheet 3)	S_y	655	MPa	Linked	Linked (single tempering point)	
B8	Working stress (from B6)	σ	120	MPa	Calc	Linked to block B6	
B8	Factor of safety	n	5.45833333333334	–	Calc		
B8	Check: reciprocal route	n_chk	5.45833333333334	–	Check		
B8	Verification tolerance (relative)	tol	0.005	–	Input	Pass if |check − result| / result ≤ tol	
B8	Independent check: |chk − n| / n	Δrel	0.000%	–	Status	Relative difference between primary and independent route	Verified
B9	**Thermal stress (fully restrained)**					**Equation: σ = E·α·ΔT**	
	Assumptions / applicability					Full restraint, uniform temperature, linear-elastic, constant E and α, σ below yield; real restraint is partial.	
B9	E, AISI 1045 HR (Sheet 3)	E	205.0	GPa	Linked	Linked	
B9	CTE, AISI 1045 HR (Sheet 3)	α	11.9	µm/(m·K)	Linked	Linked	
B9	Temperature rise	ΔT	50	K	Input	Assumed	
B9	Thermal stress	σ	121.975	MPa	Calc		
B9	Check: SI route	σ_chk	121.975	MPa	Check		
B9	Verification tolerance (relative)	tol	0.005	–	Input	Pass if |check − result| / result ≤ tol	
B9	Independent check: |chk − sg| / sg	Δrel	0.000%	–	Status	Relative difference between primary and independent route	Verified
B10	**Fatigue design parameter — modified Goodman factor of safety**					**Equation: 1/n = σa/Se + σm/Sut ; Se ≈ 0.5·Sut (steel, Sut ≤ ~1400 MPa, rotating-beam specimen)**	
	Assumptions / applicability					Infinite-life, proportional loading, uniaxial; real Se needs Marin factors (surface, size, load, reliability, notch). Illustrative inputs.	
B10	Alternating stress	σa	100	MPa	Input	Assumed	
B10	Mean stress	σm	150	MPa	Input	Assumed	
B10	Ultimate strength	Sut	600	MPa	Input	Assumed	
B10	Endurance limit estimate	Se	300	MPa	Calc	0.5 factor is a steel rule of thumb	
B10	Goodman factor of safety	n	1.71428571428571	–	Calc		
B10	Expected value on Goodman line	1	1	–	Calc	At n·σ the point lies on the line	
B10	Check: n·σa/Se + n·σm/Sut	chk	1	–	Check	Must equal 1	
B10	Verification tolerance (relative)	tol	0.005	–	Input	Pass if |check − result| / result ≤ tol	
B10	Independent check: |exp − chk| / chk	Δrel	0.000%	–	Status	Relative difference between primary and independent route	Verified
B11	**Grain-size strengthening (Hall–Petch)**					**Equation: σy = σ0 + ky / √d**	
	Assumptions / applicability					Valid for typical polycrystalline grains (≈ µm–tens of µm); breaks down at nanoscale. σ0, ky below are ASSUMED illustrative values (order of magnitude for mild steel), not fitted data.	
B11	Friction stress	σ0	70	MPa	Input	Assumed illustrative	
B11	Hall–Petch slope	ky	0.6	MPa·m^½	Input	Assumed illustrative	
B11	Grain diameter	d	2.000E-05	m	Input	20 µm assumed	
B11	Yield strength	σy	204.164078649987	MPa	Calc		
B11	Check: mm units	σy_chk	204.164078649987	MPa	Check	ky converted to MPa·mm^½	
B11	Verification tolerance (relative)	tol	0.005	–	Input	Pass if |check − result| / result ≤ tol	
B11	Independent check: |chk − sy| / sy	Δrel	0.000%	–	Status	Relative difference between primary and independent route	Verified
B12	**ASTM E112 grain size number**					**Equation: n = 2^(G−1) grains per in² at 100× ; real grains/mm² = n ÷ (25.4/100)² ; d ≈ √(area) (square-grain approx.)**	
	Assumptions / applicability					Single-phase equiaxed structure; planimetric definition.	
B12	ASTM grain size number	G	8	–	Input	Assumed	

Block	Quantity / step	Symbol	Value	Unit	Type	Equation / assumptions / note	Status
B12	mm per inch	in	25.4	mm	Input		
B12	Magnification	M	100	×	Input		
B12	Grains per in² at 100×	n	128	1/in²	Calc		
B12	Real area represented by 1 in² at 100×	a	0.064516	mm²	Calc		
B12	Grains per mm²	N	1984.00396800794	1/mm²	Calc		
B12	Mean grain size (square-grain approx.)	d	22.4506403026729	µm	Calc		
B12	ASTM tabulated mean grain diameter, G=8 (recalled)	d_ref	22.5	µm	Input	Recalled table value — not re-fetched	
B12	Verification tolerance (relative)	tol	0.02	–	Input	Pass if |check − result| / result ≤ tol	
B12	Independent check: |chk − dd| / dd	Δrel	0.220%	–	Status	Relative difference between primary and independent route	Verified
B13	**Carbon equivalent (IIW)**					**Equation: CE = C + Mn/6 + (Cr+Mo+V)/5 + (Ni+Cu)/15 (wt%)**	
	Assumptions / applicability					Weldability screening only; preheat depends on thickness, hydrogen level, restraint and code. Illustrative composition.	
B13	Carbon	C	0.2	wt%	Input	Assumed	
B13	Manganese	Mn	0.9	wt%	Input		
B13	Chromium	Cr	0.05	wt%	Input		
B13	Molybdenum	Mo	0	wt%	Input		
B13	Vanadium	V	0	wt%	Input		
B13	Nickel	Ni	0.05	wt%	Input		
B13	Copper	Cu	0.05	wt%	Input		
B13	Carbon equivalent	CE	0.36666666666667	wt%	Calc		
B13	Check: term by term	CE_chk	0.36666666666667	wt%	Check		
B13	Verification tolerance (relative)	tol	0.005	–	Input	Pass if |check − result| / result ≤ tol	
B13	Independent check: |chk − CE| / CE	Δrel	0.000%	–	Status	Relative difference between primary and independent route	Verified
B14	**Pitting resistance equivalent number (PREN)**					**Equation: PREN = %Cr + 3.3·%Mo + 16·%N**	
	Assumptions / applicability					Ranking index only (no unique N factor; Mn/S/W variants exist); not a service guarantee. Illustrative mid-range 2205 composition.	
B14	Chromium	Cr	22.5	wt%	Input	Assumed	
B14	Molybdenum	Mo	3.2	wt%	Input		
B14	Nitrogen	N	0.17	wt%	Input		
B14	PREN	PREN	35.78	–	Calc		
B14	Check: fraction form	PREN_chk	35.78	–	Check		
B14	Verification tolerance (relative)	tol	0.005	–	Input	Pass if |check − result| / result ≤ tol	
B14	Independent check: |chk − PREN| / PREN	Δrel	0.000%	–	Status	Relative difference between primary and independent route	Verified
B15	**Larson–Miller creep parameter**					**Equation: LMP = T·(C + log10 t) ; T in K, t in h**	
	Assumptions / applicability					C ≈ 20 is a typical default; alloy-specific C and the stress–LMP master curve must come from creep test data (ASTM E139).	
B15	Temperature	T	600	°C	Input	Assumed	
B15	Larson–Miller constant	C	20	–	Input	Assumed typical	
B15	Rupture time	t	1000	h	Input	Assumed	
B15	Temperature	T	873.15	K	Calc		
B15	Larson–Miller parameter	LMP	20082.45	K	Calc		
B15	Check: invert for t	t_chk	1000	h	Check	Should return t	
B15	Verification tolerance (relative)	tol	0.005	–	Input	Pass if |check − result| / result ≤ tol	

Block	Quantity / step	Symbol	Value	Unit	Type	Equation / assumptions / note	Status
B15	Independent check: |t − chk| / chk	Δrel	0.000%	–	Status	Relative difference between primary and independent route	Verified
B16	**Hardness–UTS approximation (steels)**					**Equation: UTS ≈ 3.45 × HBW (MPa)**	
	Assumptions / applicability					Carbon & low-alloy steels only (roughly 150–450 HBW); not for austenitic SS, cold-worked or cast irons; ±~10% scatter; for estimates, not acceptance.	
B16	Brinell hardness	HBW	200	HBW	Input	Assumed	
B16	Estimated UTS	UTS	690	MPa	Calc		
B16	Check: 500·HBW psi → MPa	UTS_chk	689.476	MPa	Check	psi→MPa factor 0.00689476	
B16	Verification tolerance (relative)	tol	0.005	–	Input	Pass if |check − result| / result ≤ tol	
B16	Independent check: |chk − U| / U	Δrel	0.076%	–	Status	Relative difference between primary and independent route	Verified
B17	**Archard wear (illustrative)**					**Equation: V = K·W·s / H**	
	Assumptions / applicability					K is system-specific (from pin-on-disc type tests); no predictive use without test data. H = hardness of softer surface. Illustrative inputs.	
B17	Wear coefficient	K	1.000E-05	–	Input	Assumed	
B17	Normal load	W	100	N	Input		
B17	Sliding distance	s	1000	m	Input		
B17	Hardness of softer body	H	2000000000	Pa	Input	2 GPa assumed	
B17	Worn volume	V	5E-10	m³	Calc		
B17	Worn volume	V	0.5	mm³	Calc		
B17	Check: N/mm² route	V_chk	0.5	mm³	Check		
B17	Verification tolerance (relative)	tol	0.005	–	Input	Pass if |check − result| / result ≤ tol	
B17	Independent check: |chk − Vmm| / Vmm	Δrel	0.000%	–	Status	Relative difference between primary and independent route	Verified
B18	**Isotropic elastic constants**					**Equation: G = E / (2(1+ν)) ; K = E / (3(1−2ν))**	
	Assumptions / applicability					Isotropic, linear-elastic; invalid for CFRP laminates/textured metals.	
B18	E, AA 6061-T6 (Sheet 3)	E	68.9	GPa	Linked	Linked	
B18	ν, AA 6061-T6 (Sheet 3)	ν	0.330	–	Linked	Linked	
B18	Shear modulus	G	25.9022556390977	GPa	Calc		
B18	Bulk modulus	K	67.5490196078431	GPa	Calc		
B18	Handbook shear modulus, 6061-T6 (recalled)	G_ref	26	GPa	Input	Recalled — not re-fetched	
B18	Verification tolerance (relative)	tol	0.02	–	Input	Pass if |check − result| / result ≤ tol	
B18	Independent check: |chk − G| / G	Δrel	0.377%	–	Status	Relative difference between primary and independent route	Verified
—	Creep strain vs time	—	not computed	—	Note	Requires alloy/condition/temperature-specific creep curves (ASTM E139); no reliable test data are included, so no calculation is presented rather than an invented one.	

Sheet 6 — Testing-Method Register

Test methods are kept distinct from product specifications and acceptance criteria. Tests are NOT universally mandatory — requirements come from the drawing, PO, product spec or governing code.

ID	Test / examination	Purpose / output	Applicable materials	Specimen / sample requirement	Key limitations	ASTM reference	ISO reference	IS (BIS) reference	Method vs specification / when appropriate	Reference verification status
T01	Tensile test	Yield/proof, UTS, elongation, reduction of area, modulus (indirect)	Metals (all)	Machined or full-section specimen per method, e.g. round Ø12.5 mm × 50 mm gauge (E8 standard)	Needs calibrated extensometer for modulus; result depends on orientation, rate, temperature	ASTM E8/E8M	ISO 6892-1	IS 1608 (Part 1)	Method standard — acceptance limits come from product spec/PO	Conditionally verified
T02	Tensile test (plastics)	Tensile strength/modulus/elongation of plastics	Polymers	Dumbbell (Type I etc.), conditioned	Rate/moisture sensitive	ASTM D638	ISO 527-1/-2	—	Method standard; datasheet states conditioning	Conditionally verified
T03	Tensile test (composites)	In-plane tensile strength/modulus of polymer-matrix composites	CFRP/GFRP	Flat coupon with tabs	Gripping failures; layup-specific	ASTM D3039	ISO 527-4/-5	—	Method standard	Conditionally verified
T04	Brinell hardness	HBW — bulk hardness of soft/mid-hard metals, castings	Steels, cast iron, non-ferrous	Flat surface; thickness ≥10× indent depth; ball size/force per material	Not for thin/hard (>650 HBW) parts	ASTM E10	ISO 6506-1	IS 1500 (Part 1)	Method standard; acceptance hardness from spec	Conditionally verified
T05	Rockwell hardness	HRB/HRC etc. — rapid hardness	Steels, Al, Cu alloys	Smooth surface; thickness ≥10× depth; scale chosen by material	Scale conversion is approximate	ASTM E18	ISO 6508-1	IS 1586 (Part 1)	Method standard	Conditionally verified
T06	Vickers hardness	HV — hardness of thin sections, case depth profiles	All metals, ceramics (micro)	Polished surface; diagonals measured	Microhardness scatter; load effects	ASTM E92 (macro), E384 (micro)	ISO 6507-1	IS 1501 (Part 1) (verify)	Method standard	Unverified
T07	Hardness conversion	Approximate conversion between scales/UTS	Steels	—	Approximate only	ASTM E140; A370	ISO 18265	—	Informative	Conditionally verified
T08	Charpy impact	Absorbed energy (J), transition behaviour	Steels (BCC), some others	10×10×55 mm V-notch typical; temperature-controlled; ≥3 specimens typical	Subsize specimens need correction; not a design property by itself	ASTM E23	ISO 148-1	IS 1757 (Part 1)	Method standard; temperature/energy from spec	Conditionally verified
T09	Izod/Charpy for plastics	Impact resistance of plastics	Polymers	Notched specimens	Notch sensitivity; not directly comparable to metals	ASTM D256; D6110 (Charpy)	ISO 180 / 179	—	Method standard	Conditionally verified
T10	Bend test	Ductility/ weld soundness	Plate, welds	Guided bend per method	Qualitative pass/fail	ASTM E290	ISO 7438	IS 1599 (verify)	Method standard	Unverified
T11	Density measurement	Bulk density; porosity indication	Metals, ceramics, polymers	Archimedes or geometric	Open porosity affects result	ASTM B311 (hardmetal), D792 (plastics), C373 (ceramics)	ISO 1183-1, ISO 3369	—	Method standards for different material classes	Conditionally verified
T12	Chemical composition (OES/XRF)	Chemistry for grade verification	Metals	Representative sample/surface prep	XRF cannot measure C; OES destructive	ASTM E415 (steel OES), E1086 (stainless OES), E1476 (identification guide)	ISO 14284 (sampling)	IS 228 series (verify)	Method standards; ranges from product spec	Unverified
T13	Interstitial/gas analysis (C, S, N, O, H)	C/S by combustion; O/N/H by fusion	Steels, Ti, Ni	Chip/pin sample	Needs calibration standards	ASTM E1019	ISO 15350	—	Method standard	Conditionally verified
T14	Metallographic preparation	Cutting, mounting, grinding, polishing, etching	All metals	Representative section; orientation stated	Preparation artefacts	ASTM E3 (guide)	ISO 4967 (inclusions) related	—	Guide, not acceptance criterion	Conditionally verified
T15	Microetching / microstructure	Reveal phases and grain structure	All metals	Polished specimen + etchant	Etchant choice crucial	ASTM E407	—	—	Method standard	Conditionally verified
T16	Grain-size determination	ASTM grain size number G; intercept/planimetric	Single-phase metals; steels	Etched polished section	Method differs for duplex/lamellar structures	ASTM E112	ISO 643	IS 4748 (verify)	Method standard	Unverified
T17	Inclusion rating	Steel cleanliness	Bearing/gear/spring steels	Longitudinal section	Area sampled is tiny	ASTM E45	ISO 4967	—	Method; limits from spec/PO	Conditionally verified
T18	Case depth / decarburisation	Effective case depth, decarb depth	Case-hardened/carbon steels	Microhardness traverse or microstructure	Definition of ECD (e.g. 550 HV) must be stated	ASTM E1077, E384	ISO 2639, ISO 3754	—	Method; depth requirement from drawing	Conditionally verified
T19	Corrosion immersion test	Mass loss/ corrosion rate	Metals	Cleaned, weighed coupons	Lab ≠ service	ASTM G31; G1 (cleaning)	ISO 8407	—	Method standard	Conditionally verified
T20	Intergranular corrosion (austenitic SS)	Detect sensitisation	Austenitic SS	Per practice (A–F)	Practice selection depends on grade	ASTM A262	ISO 3651-2	—	Applicable only when specified/required	Conditionally verified
T21	Pitting/crevice corrosion (SS)	Ranking resistance by FeCl3 test	Stainless & Ni alloys	Per method	Ranking, not life prediction	ASTM G48	ISO 17475 (guidance)	—	Applicable when pitting is a risk	Conditionally verified
T22	Salt-spray (fog)	Apparatus practice for accelerated corrosion screening of coatings/materials	Coated and bare metals	Panels in neutral salt fog	Poor correlation to service life; no inherent pass/fail	ASTM B117	ISO 9227	—	Operating practice; acceptance from coating/product spec	Conditionally verified
T23	Liquid penetrant testing	Surface-breaking defects in non-porous materials	Metals (non-porous)	Clean surface; method/sensitivity per procedure	Not for porous surfaces; surface-open flaws only	ASTM E165/E165M; E1417	ISO 3452-1	—	Practice; acceptance from code/spec	Conditionally verified
T24	Magnetic particle testing	Surface/near-surface flaws in ferromagnetic materials	Carbon/low-alloy steels, martensitic SS	Magnetisable part	Not for austenitic SS, Al, Cu, Ti	ASTM E1444/E1444M	ISO 9934-1	—	Practice; acceptance from code/spec	Conditionally verified
T25	Ultrasonic testing — welds	Sub-surface weld flaws by contact UT	Welded joints	Calibrated probes/reference blocks	Operator/ procedure dependent	ASTM E164 (status to verify)	ISO 17640	—	Practice; acceptance from code	Unverified
T26	Ultrasonic testing — straight beam (bar/plate/forging)	Internal discontinuities	Wrought products	Contact straight-beam procedure	Orientation sensitivity	ASTM E114 (verify)	ISO 16810 (general)	—	Practice; acceptance from product spec	Unverified
T27	Radiographic testing	Volumetric internal flaws (porosity, cracks, inclusions)	Castings, welds	Film/digital with image-quality indicators	Radiation safety; planar flaw orientation	ASTM E1742/E1742M	ISO 17636-1	—	Practice; acceptance from code/casting spec	Conditionally verified
T28	Creep / stress-rupture	Creep rate, rupture time	High-temperature alloys	Constant load/temperature specimens	Long duration; extrapolation uncertainty	ASTM E139	ISO 204	—	Applicable when service ≥0.4 Tm	Conditionally verified
T29	Fatigue (force-controlled)	S–N behaviour	Metals	Smooth/notched axial specimens	Surface finish, mean stress, size effects	ASTM E466	ISO 1099	—	Applicable for cyclic service	Conditionally verified
T30	Fatigue (strain-controlled)	Low-cycle fatigue	Metals	Axial strain-controlled specimens	Complex; strain measurement	ASTM E606/E606M	ISO 12106	—	Applicable for LCF	Conditionally verified
T31	Fatigue crack growth	da/dN vs ΔK	Metals	Pre-cracked CT/ M(T) specimens	Lab-to-service transfer	ASTM E647	ISO 12108	—	Damage-tolerance design	Conditionally verified
T32	Rotating-bending fatigue	Endurance limit estimation	Steels, Al	Rotating beam specimen	Simplified loading	—	ISO 1143	—	Method standard	Conditionally verified
T33	Wear testing	Wear rate/friction	Metals, polymers, ceramics	Pin-on-disc etc.	System-dependent; no universal wear constant	ASTM G99 (pin-on-disc), G65 (dry sand/rubber wheel), G133	ISO 20808 (ceramics, verify)	—	Test method; results only valid for tested pairing	Unverified
T34	Thermal conductivity / diffusivity	k, α_th	All	Disc (laser flash) or bar (comparative)	Needs density & Cp	ASTM E1461 (laser flash), E1225, C177	ISO 22007-4	—	Method standard	Conditionally verified
T35	Thermal expansion	CTE, dilatometry	All	Rod/pushrod or TMA specimen	Temperature range must be stated	ASTM E228, E831	ISO 7991 (glass) / ISO 11359	—	Method standard	Conditionally verified
T36	Electrical resistivity / conductivity	Resistivity (Ω·m), %IACS	Conductors	Wire/strip with 4-point measurement	Temperature correction needed	ASTM B193	IEC 60468 (verify)	—	Method standard	Unverified
T37	Elastic modulus (dynamic)	E, G, ν by impulse/resonance	Metals, ceramics	Bar/plate specimen	Frequency/temperature effects	ASTM E1876, E111	ISO 12680-1 (verify)	—	Method standard	Unverified
T38	Fracture toughness	KIC, J, CTOD	Metals	Pre-cracked CT/SENB	Thickness/plane-strain validity limits	ASTM E399, E1820	ISO 12135	—	Applicable for damage-tolerance/critical parts	Conditionally verified
T39	Flexural strength (ceramics)	Modulus of rupture at RT	Advanced ceramics	Rectangular bars, 3/4-pt	Weibull scatter; surface-flaw dominated	ASTM C1161	ISO 14704	—	Method standard	Conditionally verified
T40	Hardness/KIC (ceramics)	Vickers hardness; fracture toughness	Ceramics	Polished surface/SEVNB	Indentation toughness unreliable	ASTM C1327, C1421	ISO 14705, ISO 23146	—	Method standard	Conditionally verified
T41	Composite flexural / short-beam	Flexural and ILSS	CFRP/GFRP	Rectangular coupon	Short-beam strength is not true shear strength	ASTM D7264, D2344	ISO 14125, ISO 14130	—	Method standard	Conditionally verified
T42	Composite fibre volume & voids	Fibre fraction, void content	Composites	Cut samples	Method choice by fibre/matrix	ASTM D3171, D2734	ISO 14127	—	Method standard	Conditionally verified
T43	DSC / transformation temperature	Tg, Tm, crystallinity; NiTi Af	Polymers, NiTi	Small sample	Heating-rate effects	ASTM E1356, E794, F2004	ISO 11357	—	Method standard	Conditionally verified
T44	Water absorption	Moisture uptake	Polymers	Dried/conditioned specimen	Time/temperature dependent	ASTM D570	ISO 62	—	Method standard	Conditionally verified

Sheet 7 — Standards Verification (14 supplied + supporting references)

Full standard texts are paywalled and no network was available: titles/editions are recalled and must be confirmed in the ASTM/BIS catalogue. Nothing is marked Verified.

ID	Original designation	Corrected designation	Official title (recalled)	Organisation	Document type	Revision status	Latest edition (recalled)	Scope / applicability	Supplied-claim issue	Official source (catalogue home)	Verification status	Notes
S01	ASTM B117	ASTM B117	Standard Practice for Operating Salt Spray (Fog) Apparatus	ASTM	Practice (apparatus operation)	Active (recalled)	B117-19 (recalled)	Operation of salt-fog apparatus for corrosion screening	Described as 'salt-spray testing' — it is an apparatus practice; it defines no pass/fail and does not predict service life	https://www.astm.org/	Conditionally verified	Acceptance criteria come from the coating/product spec.
S02	ASTM E165/E165M	ASTM E165/E165M (also see E1417/E1417M)	Standard Practice for Liquid Penetrant Testing for General Industry (older title used 'Examination')	ASTM	Practice	Active (recalled); title wording changed over revisions	Not confirmed	Liquid penetrant testing for general industry (surface-open flaws, non-porous materials)	Title wording 'examination' may be an older title — confirm current wording	https://www.astm.org/	Conditionally verified	Full text not examined.
S03	ASTM E1444/E1444M	ASTM E1444/E1444M	Standard Practice for Magnetic Particle Testing (older: 'Examination')	ASTM	Practice	Active (recalled)	Not confirmed	MT of ferromagnetic materials	Title wording may differ by edition	https://www.astm.org/	Conditionally verified	Full text not examined.
S04	ASTM E164	ASTM E164 (status unverified)	Standard Practice for Contact Ultrasonic Testing of Weldments	ASTM	Practice	UNVERIFIED — current/withdrawn status could not be confirmed	Not confirmed	Contact UT of weldments	User flagged 'subject to scope verification' — status itself uncertain	https://www.astm.org/	Unverified	Check ASTM catalogue before citing; ISO 17640 is the ISO counterpart.
S05	ASTM E1742/E1742M	ASTM E1742/E1742M	Standard Practice for Radiographic Examination	ASTM	Practice	Active (recalled)	Not confirmed	Radiographic examination of metallic/non-metallic materials	—	https://www.astm.org/	Conditionally verified	Full text not examined.
S06	ASTM E3	ASTM E3	Standard Guide for Preparation of Metallographic Specimens	ASTM	Guide	Active (recalled)	E3-11(2017) (recalled)	Specimen preparation for metallography	It is a Guide (not a test method)	https://www.astm.org/	Conditionally verified	Guides give no acceptance criteria.
S07	IS 1608 (Part 1)	IS 1608 (Part 1)	Metallic materials — Tensile testing: Part 1 Method of test at room temperature	BIS	Test method	Active (recalled)	2022 (recalled)	Room-temperature tensile test (aligned with ISO 6892-1)	Part number consistent; elevated temperature is a separate part	https://www.bis.gov.in/	Conditionally verified	Edition year not independently confirmed.
S08	IS 1500 (Part 1)	IS 1500 (Part 1)	Metallic materials — Brinell hardness test: Part 1 Test method	BIS	Test method	Active (recalled)	2019 (recalled)	Brinell hardness (aligned with ISO 6506-1)	—	https://www.bis.gov.in/	Conditionally verified	Edition year not independently confirmed.
S09	IS 1586 (Part 1)	IS 1586 (Part 1)	Metallic materials — Rockwell hardness test: Part 1 Test method	BIS	Test method	Active (recalled)	2018 (recalled)	Rockwell hardness (aligned with ISO 6508-1)	—	https://www.bis.gov.in/	Conditionally verified	Edition year not independently confirmed.
S10	IS 1757 (Part 1)	IS 1757 (Part 1)	Metallic materials — Charpy pendulum impact test: Part 1 Test method	BIS	Test method	Active (recalled)	2014 (recalled)	Charpy pendulum impact (aligned with ISO 148-1)	—	https://www.bis.gov.in/	Conditionally verified	Edition year not independently confirmed.
S11	IS 2062	IS 2062:2011 (verify later amendments)	Hot rolled medium and high tensile structural steel — Specification	BIS	Product specification	Active (recalled)	2011 (recalled)	Grades E250–E650 with quality designations A, BR, B0, C	Grade designation alone (E250/E350) is incomplete without quality designation	https://www.bis.gov.in/	Conditionally verified	A product spec, not a test method.
S12	IS 210	IS 210:2009	Grey iron castings — Specification	BIS	Product specification	Active (recalled)	2009 (recalled)	Grey iron castings FG 150–FG 400	Product spec — tests reference other IS methods	https://www.bis.gov.in/	Conditionally verified	Edition year not independently confirmed.
S13	IS 1865	IS 1865 (latest revision unverified)	Iron castings with spheroidal or nodular graphite — Specification	BIS	Product specification	Revision status unverified	1991 reaffirmed (recalled); newer revision possible	Spheroidal graphite iron castings	May have been revised — confirm in BIS catalogue	https://www.bis.gov.in/	Unverified	Do not cite edition year until verified.
S14	IS 6911	IS 6911:2017	Stainless steel plate, sheet and strip — Specification	BIS	Product specification	Active (recalled)	2017 (recalled)	Stainless plate, sheet and strip only (not bar/tube)	Scope limited to flat products	https://www.bis.gov.in/	Conditionally verified	Edition year not independently confirmed.
S15	ASTM E8/E8M	ASTM E8/E8M	Standard Test Methods for Tension Testing of Metallic Materials	ASTM	Test method	Active (recalled)	E8/E8M-22 (recalled)	Metal tensile testing	—	https://www.astm.org/	Conditionally verified	Cited for tensile tests.
S16	ASTM E10	ASTM E10	Standard Test Method for Brinell Hardness of Metallic Materials	ASTM	Test method	Active (recalled)	Not confirmed	Brinell hardness	—	https://www.astm.org/	Conditionally verified	
S17	ASTM E18	ASTM E18	Standard Test Methods for Rockwell Hardness of Metallic Materials	ASTM	Test method	Active (recalled)	Not confirmed	Rockwell hardness	—	https://www.astm.org/	Conditionally verified	
S18	ASTM E23	ASTM E23	Standard Test Methods for Notched Bar Impact Testing of Metallic Materials	ASTM	Test method	Active (recalled)	Not confirmed	Charpy/Izod for metals	—	https://www.astm.org/	Conditionally verified	
S19	ASTM E112	ASTM E112	Standard Test Methods for Determining Average Grain Size	ASTM	Test method	Active (recalled)	Not confirmed	Grain size by comparison/intercept/planimetric	—	https://www.astm.org/	Conditionally verified	Used in grain-structure sheet.
S20	ASTM E45	ASTM E45	Standard Test Methods for Determining the Inclusion Content of Steel	ASTM	Test method	Active (recalled)	Not confirmed	Inclusion rating	—	https://www.astm.org/	Conditionally verified	
S21	ASTM A240/A240M	ASTM A240/A240M	Standard Specification for Chromium and Chromium-Nickel Stainless Steel Plate, Sheet, and Strip for Pressure Vessels and for General Applications	ASTM	Product specification	Active (recalled)	Not confirmed	Stainless plate/sheet/strip	—	https://www.astm.org/	Conditionally verified	Property minima used in Sheet 3.
S22	ASTM A262	ASTM A262	Standard Practices for Detecting Susceptibility to Intergranular Attack in Austenitic Stainless Steels	ASTM	Practice	Active (recalled)	Not confirmed	IGC screening (practices A–F)	—	https://www.astm.org/	Conditionally verified	
S23	ASTM A29/A29M	ASTM A29/A29M	Standard Specification for Steel Bars, Carbon and Alloy, Hot-Wrought, General Requirements	ASTM	Product specification (general requirements)	Active (recalled)	Not confirmed	Chemistry/tolerances for carbon & alloy bar	—	https://www.astm.org/	Conditionally verified	
S24	ASTM D638	ASTM D638	Standard Test Method for Tensile Properties of Plastics	ASTM	Test method	Active (recalled)	Not confirmed	Plastics tensile	—	https://www.astm.org/	Conditionally verified	
S25	ASTM C1161	ASTM C1161	Standard Test Method for Flexural Strength of Advanced Ceramics at Ambient Temperature	ASTM	Test method	Active (recalled)	Not confirmed	Ceramic flexure	—	https://www.astm.org/	Conditionally verified	
S26	ASTM D3039/D3039M	ASTM D3039/D3039M	Standard Test Method for Tensile Properties of Polymer Matrix Composite Materials	ASTM	Test method	Active (recalled)	Not confirmed	Composite tension	—	https://www.astm.org/	Conditionally verified	
S27	ISO 6892-1	ISO 6892-1	Metallic materials — Tensile testing — Part 1: Method of test at room temperature	ISO	Test method	Active (recalled)	Not confirmed	Counterpart of IS 1608 (Part 1)	—	https://www.iso.org/	Conditionally verified	

Sheet 8 — Errors and Corrections

Original claim → detected issue → correction. Status shows the verification state of the correction itself.

ID	Source	Original claim	Detected issue / correction	Technical reasoning	Downstream effects	Evidence	Status
E01	Upload rows 28–30	EN8, EN19, EN24 listed as grades	Legacy BS 970 designations (080M40, 709M40/708M40, 817M40); BS 970 withdrawn/replaced by BS EN 10083	Designations obsolete; modern grades similar but chemistry/test requirements differ	No properties assigned to EN8/EN19/EN24; use C45/42CrMo4/34CrNiMo6 with EN 10083 specs	Recalled standards history (not re-fetched)	Incorrect
E02	Upload row 9	'AISI 1045 / C45' implies equivalence	Treat as similar, not identical	AISI/SAE limits (Mn, P, S) differ from EN 10083-2 C45 ranges	Properties in Sheet 3 recorded for AISI 1045 only	Recalled chemistry limits	Conditionally verified
E03	Upload row 17	'AISI 4140 / 42CrMo4'	Similar, not identical (Cr 0.80–1.10 vs 0.90–1.20; Mo 0.15–0.25 vs 0.15–0.30; Mn 0.75–1.00 vs 0.60–0.90, recalled)	Different chemistry windows can change hardenability	Properties recorded for AISI 4140 only	Recalled chemistry limits	Conditionally verified
E04	Upload row 52	'AISI 52100 / 100Cr6'	Close, not identical; verify via ASTM A295 vs ISO 683-17	Cr/Mn/Si limits differ slightly	Bearing-grade cleanliness must be specified separately	Recalled	Conditionally verified
E05	Upload rows 31–44	'SS 201/304/…' naming	Use ASTM A240 Type/UNS/EN 10088/IS 6911 designation	'SS xxx' is not a standard designation system	Prevents 201/304 mix-ups in procurement	Recalled UNS numbers	Conditionally verified
E06	Upload rows 23, 24, 55	40Cr, 20CrMo, 65Mn	Chinese GB/T designations; no BIS/ASTM equivalence asserted	Equivalence requires published cross-reference	No numerical properties recorded	Recalled	Conditionally verified
E07	Upload rows 91–95, 122, 125, 126, 131, 132	Trade names (Inconel, Monel, Hastelloy, Nitinol, Hardox)	Use UNS/ASTM designations; Hardox is SSAB-proprietary	Trade names are not standards	Hardox properties = manufacturer datasheet	Recalled	Conditionally verified
E08	Upload rows 111–121, 123, 124, 129	Under-specified materials (PET-P, alumina, SiC, ZrO2, Si3N4, CFRP, GFRP, aramid, CoCr, graphite, CuBe)	Specify exact grade/class (purity, sinter route, layup, ASTM F75/F1537/F90, etc.)	Properties vary several-fold	Only indicative ranges recorded, marked Low confidence	Judgement	Unverified
E09	Upload row 43	Duplex 2205	UNS S32205 and older S31803 both exist (S32205 has tighter N/Cr/Mo limits)	Two UNS numbers cause confusion	Property row uses S32205 A240 minima	Recalled	Conditionally verified
E10	Upload row 44	17-4 PH without condition	Condition H900–H1150 required	Strength/toughness vary strongly with ageing	Property row recorded for H1025 only	Recalled	Conditionally verified
E11	Upload rows 12–13	IS 2062 E250/E350 without quality designation	Add A / BR / B0 / C and thickness band	Impact and test options depend on designation	Properties listed as minima for ≤20 mm only	Recalled	Conditionally verified
E12	Master prompt §5	ASTM E3 described as 'preparation of metallographic specimens'	E3 is a Guide	Guides give recommendations, not acceptance criteria	Reflected in Standards sheet	Recalled	Conditionally verified
E13	Master prompt §5	ASTM B117 as 'salt-spray testing'	It is an apparatus operating practice	Does not define pass/fail or predict life	Never use B117 hours alone as a service-life claim	Recalled	Conditionally verified
E14	Master prompt §5	ASTM E165/E165M & E1444/E1444M titles 'examination'	Current titles use 'testing' (older editions 'examination') — confirm	Title wording varies by edition	Titles flagged Conditionally verified	Recalled	Conditionally verified
E15	Master prompt §5	ASTM E164 status	Current/withdrawn status could not be confirmed	No authoritative access	Marked Unverified	—	Unverified
E16	Master prompt §5	IS 1865 edition not known	Edition/revision status unverified	BIS catalogue not accessible	Marked Unverified	—	Unverified
E17	Upload test notes	Tests listed without standards for metals	Added ASTM/ISO/IS references in Testing Methods	Improves traceability	See Sheet 'Testing Methods'	—	Conditionally verified
E18	Upload (scope)	No crystal structure/grain/microstructure fields	Added qualified crystal structure, phases and grain notes	Requested by user; avoids family-wide single structure	See 'Grain & Crystal Structure'	—	Conditionally verified
E19	Upload (scope)	AM, super-duplex, PH, Indian alloy steels missing	Added gap-list candidates (not counted in coverage)	Important for robotics/aerospace/Indian procurement	Bottom of Master Material Database	—	Unverified
E20	Master prompt §2	CTE unit µm/(m·K) = µm/(m·°C)	Correct for temperature differences	Kelvin and Celsius intervals are equal	—	Unit analysis	Verified
E21	Master prompt §3	Supplied formulas (ρ, ΔL, R, σ/ρ, E/ρ)	Dimensionally correct	Checked in worksheet with independent recalculation	See Formulas sheet	Computed in workbook	Verified
E22	Master prompt §8	Target 90–95% verified coverage	Not achievable in this session: no network/authoritative source access	Strict coverage counts only items checked against official documents	Coverage reported as strict and conditional	Executive Summary	Unverified

Sheet 9 — Industry Considerations (reviewer's additions)

Practical issues that decide real-world material success. Numbers are approximate/typical and marked for verification; this sheet is advisory and excluded from verification coverage.

ID	Topic	Category	Materials affected	What happens / why it matters	Typical numbers / limits (approx.)	Control / mitigation	Relevant standard / test (verify)	Status
I01	Sensitisation / intergranular corrosion	Corrosion	Austenitic SS (304, 316, 321 vs L grades)	Cr carbides precipitate at grain boundaries when held ≈450–850 °C, depleting adjacent Cr	Risk rises with carbon and exposure time	Use L or stabilised (321/347) grades; solution anneal after welding if needed	ASTM A262 (verify practice letter for service)	Advisory (not counted)
I02	Chloride stress-corrosion cracking & pitting	Corrosion	304/316, duplex, 17-4 PH	Tensile stress + chlorides + temperature causes cracking; pitting ranks by PREN	Austenitic SCC concern generally above ≈50–60 °C in chlorides	Choose duplex/super-duplex/Ni alloys/Ti; reduce stress; coatings	ASTM G48 (pitting/crevice), G36 (verify)	Advisory (not counted)
I03	Sigma phase / 475 °C embrittlement	Metallurgy	Duplex, ferritic, 310S	Intermetallic σ (≈600–1000 °C) and α′ (≈400–540 °C) cause toughness loss	Duplex service limit commonly ≈250–300 °C	Respect temperature limits; control weld heat input; solution anneal	Charpy; ASTM A923 (verify)	Advisory (not counted)
I04	Ductile-to-brittle transition (BCC)	Fracture	Carbon/low-alloy steels, ferritic SS, Mo, W	BCC metals lose toughness at low temperature/high strain rate; FCC (Al, Cu, austenitic, Ni) do not	Transition temperature depends on grain size, composition, thickness	Specify impact test temperature; use fine-grain/Ni steels or FCC alloys for cold service	ASTM E23 / IS 1757 (Part 1) / ISO 148-1	Advisory (not counted)
I05	Hydrogen embrittlement	Environment	High-strength steels, 17-4 PH H900, Ti, some Ni alloys	Atomic H from plating/pickling/cathodic protection causes delayed brittle fracture	Susceptibility rises above ≈1000–1200 MPa (≈ >32–35 HRC)	Baking after plating (e.g. ≈190–220 °C for hours per ASTM B850, verify); limit hardness	ASTM F519 (verify), B850	Advisory (not counted)
I06	Temper & tempered-martensite embrittlement	Heat treatment	4130/4140/4340, Cr-Ni-Mo steels	Tempering ≈250–400 °C (TME) or slow-cooling through ≈375–575 °C (temper embrittlement from P, Sn, Sb, As) reduces toughness	Mo additions reduce susceptibility	Avoid embrittling temper ranges; cleaner steel; rapid cooling after temper	Charpy transition curves	Advisory (not counted)
I07	Weld cold cracking & carbon equivalent	Welding	Medium/high-carbon & alloy steels, HSLA	Hard HAZ + hydrogen + residual stress → cracking	IIW CE > ≈0.4 raises preheat need (thickness & process dependent)	Preheat, low-hydrogen consumables, PWHT, interpass control	EN 1011-2, AWS D1.1 (verify)	Advisory (not counted)
I08	Weld HAZ softening of aluminium	Welding	6061-T6, 7xxx, 2xxx, work-hardened 5xxx	Precipitates dissolve/coarsen in HAZ; T6 strength can fall by >40% locally	Design codes apply HAZ reduction factors	Design for HAZ strength; post-weld ageing where possible; choose 5xxx for welded structures	Weld procedure qualification; hardness traverse	Advisory (not counted)
I09	Stress-corrosion in 7xxx and 2xxx aluminium	Corrosion	7075-T6, 2024-T3	SCC in short-transverse grain direction; exfoliation	T73/T76 overaged tempers resist SCC with strength penalty	Avoid sustained tensile stress in ST direction; select T73	ASTM G47, G34 (verify)	Advisory (not counted)
I10	Galvanic corrosion	Corrosion	Mg, Al with Cu/steel/CFRP/SS	Dissimilar metals in electrolyte; carbon fibre is strongly cathodic to Al/Mg/steel	Area ratio (large cathode/small anode) is worst	Isolate (sealants/insulating layers), coatings, compatible pairs	ASTM G71 (verify)	Advisory (not counted)
I11	Cast iron behaviour	Design	Gray & ductile iron	Gray: low tensile, high compressive (≈3–4× tensile), high damping, notch-insensitive; ductile: notch-sensitive tougher	Compressive/tensile ratio approximate	Use ductile iron where shock/tension matter; check section-size effects	IS 210, IS 1865, EN 1561/1563	Advisory (not counted)
I12	Casting porosity & fatigue	Quality	Al castings, Mg die-cast, Zn die-cast, iron castings	Pores/oxides act as crack initiators; fatigue life tracks largest defect	Test bars ≠ casting properties	HIP, vacuum/low-pressure casting, X-ray/CT acceptance levels	ASTM E505/E155 (verify) radiographic references	Advisory (not counted)
I13	Inclusion cleanliness	Quality	Bearing steels, gear steels, spring steels	Oxide/sulphide inclusions initiate rolling-contact/high-cycle fatigue	Vacuum degassing/VAR/ESR improve cleanliness	Specify cleanliness rating & melt route	ASTM E45, ISO 4967	Advisory (not counted)
I14	Grain size & grain growth	Metallurgy	All polycrystalline metals	Fine grain increases yield (Hall–Petch) and toughness; abnormal grain growth after critical strain anneal or overheating	Coarse grain in Al after small cold work + anneal	Control normalising/anneal; add grain-refining elements (Al, Nb, Ti, V)	ASTM E112, E930 (verify)	Advisory (not counted)
I15	Texture & anisotropy	Design	Rolled sheet/plate, Mg, Ti, extrusions, CFRP, AM	Properties differ by direction (L/LT/ST); earing in deep-drawing	CFRP E1 ≫ E2	State test direction; use ST data for thick-section design	ASTM E8 specimen orientation notes; plate L-T/T-L	Advisory (not counted)
I16	Heat-treatment distortion & quench cracking	Manufacturing	Tool steels, alloy steels, carburised parts	Non-uniform cooling/phase changes cause warpage and cracks	Air-hardening grades (A2, D2) distort less than water-hardening	Select quench media/steel grade; stress-relieve before hardening; allow grinding stock	Dimensional inspection; MPI after grinding	Advisory (not counted)
I17	Surface engineering	Manufacturing	Steels, Al, Ti	Carburising ≈880–950 °C; gas nitriding ≈500–550 °C; induction hardening; anodising; hard chrome; PVD/DLC; shot peening improves fatigue	Process temps approximate	Specify process, case depth and hardness; control distortion and hydrogen	ISO 2639 (case depth), ASTM B117 (corrosion screening)	Advisory (not counted)
I18	Residual stress	Manufacturing	Welds, castings, forgings, AM	Residual tension lowers fatigue/SCC resistance, causes distortion after machining	Magnitude can approach yield in welds/AM	Stress relief, peening, balanced machining	ASTM E837 hole-drilling (verify), XRD	Advisory (not counted)
I19	Machinability & ratings	Manufacturing	Steels, brasses, Ti, Ni alloys	Free-machining C36000 is the 100% reference for copper alloys; AISI 1212 = 100% for steels; Ti/Ni/Inconel machine slowly with tool wear	Ratings are relative, shop-dependent	Select grade for process; use coolant/carbide tooling	Machining trials; ISO 3685 (verify)	Advisory (not counted)
I20	Galling & wear	Tribology	Austenitic SS, Ti, soft Al	Adhesive wear/seizure between similar metals	—	Dissimilar couples, hard coatings, lubrication	ASTM G98 (galling, verify)	Advisory (not counted)
I21	Creep & elevated temperature	Design	718, 625, 310S, 321, H13, Ti (>≈400 °C), Mg (>≈120 °C), Zn, polymers	Time-dependent strain at ≥0.4 Tm; rupture life by Larson–Miller	Polymers creep at RT; Zn creeps at RT	Use creep-rupture data; lower stress; larger sections	ASTM E139; ISO 204	Advisory (not counted)
I22	Oxidation & hot corrosion	High temperature	Mo, W, Ti, graphite, 310S, Ni alloys	Mo forms volatile MoO3 in air above ≈500–600 °C; graphite oxidises >≈400–500 °C	Approximate thresholds	Inert/vacuum atmosphere or coatings	Oxidation tests (ASTM G54 verify)	Advisory (not counted)
I23	Magnesium flammability	Safety	Mg alloys	Fines and chips can ignite; Mg fires need Class D extinguishers	—	Dust/chip control, coolant selection, safe storage	Facility standards (NFPA 484, verify)	Advisory (not counted)
I24	Beryllium hazard	Safety / regulatory	CuBe (C17200)	Dust/fume inhalation → chronic beryllium disease; machining controls required	Regulatory exposure limits apply (jurisdiction-specific)	Local exhaust, wet machining, regulated handling	OSHA/regulatory rules (jurisdiction)	Advisory (not counted)
I25	Lead & restricted substances	Regulatory	C36000, C93200, solders	Pb restricted in some drinking-water/RoHS/REACH contexts	Jurisdiction-specific	Lead-free alternatives; supplier declarations	RoHS/REACH (verify scope)	Advisory (not counted)
I26	Titanium specifics	Processing	Ti Gr 2, Ti-6Al-4V	Alpha case, oxygen/hydrogen pickup, galling, low thermal conductivity machining heat	—	Vacuum/inert heat treatment; remove alpha case; sharp tooling	ASTM B265/F136; ASTM E1409/E1447 (O, H analysis, verify)	Advisory (not counted)
I27	Nitinol functionality	Design	NiTi	Af temperature/superelastic window sets function; fatigue under strain amplitude; welding difficult	Af shifts by tens °C for ≈1 at% Ni change	Specify Af; DSC and bend-free-recovery tests	ASTM F2063, F2004, F2082	Advisory (not counted)
I28	Polymer moisture, creep, UV	Polymers	PA6/PA66, POM, PEEK, PTFE, UHMWPE	Moisture lowers PA stiffness/strength and changes dimensions; creep under constant load; UV degrades PC/ABS/PE	PA66 saturation moisture several % — datasheet dependent	Use conditioned data (e.g. 50% RH), reinforcement, design for creep	ASTM D570, D638, D2990 (verify)	Advisory (not counted)
I29	Ceramic design	Design	Al2O3, SiC, ZrO2, Si3N4	Strength is statistical (Weibull); contact stress and thermal shock govern failures; compression ≫ tension	Weibull modulus m typically ≈5–15 for engineering ceramics	Design for compression, avoid stress concentrators, proof test	ASTM C1161, C1239 (Weibull, verify)	Advisory (not counted)
I30	Composites damage tolerance	Design	CFRP/GFRP	Delamination and barely visible impact damage reduce compression strength; moisture lowers matrix-dominated properties	—	Damage-tolerance design, NDT (UT/thermography), conditioning tests	ASTM D7136, D7137 (verify)	Advisory (not counted)
I31	Additive manufacturing qualification	Manufacturing	L-PBF metals	Build orientation anisotropy, porosity, rough surface, powder reuse (O/N pickup), residual stress	—	Stress relief, HIP, machining critical surfaces, lot/powder traceability	ISO/ASTM 52900 terms; ASTM F3122 (verify)	Advisory (not counted)
I32	Certificates & grade mix-ups	Procurement	All; esp. SS 201/304/316, alloy bars	EN 10204 3.1/3.2 MTC gives chemistry/mechanical results; mislabelled stainless is a known risk; magnet test cannot separate 201/304/316	—	Verify MTC vs heat number; PMI (XRF/OES); independent lab tests for critical parts	EN 10204; ASTM E1476 (verify)	Advisory (not counted)
I33	Standards equivalence caution	Procurement	All	IS, ASTM, EN, ISO, JIS, GB grades are NOT equivalent unless the standards state it; chemistry, tolerances, test requirements differ	—	Name the exact standard/edition/grade/condition on the PO	BIS/ASTM/ISO catalogues	Advisory (not counted)

ID	Topic	Category	Materials affected	What happens / why it matters	Typical numbers / limits (approx.)	Control / mitigation	Relevant standard / test (verify)	Status
I34	Cost, supply & sustainability	Business	Ni, Co, Mo, Ti alloys; Al; steels	Alloy-element price volatility (Ni, Co, Mo) drives cost; recycled Al uses a small fraction of primary-Al energy (commonly cited ≈5%); steel EAF vs BF route changes embodied CO2	Not verified numerically here	Get live quotes; check recycled content/EPDs; dual-source grades	Supplier EPDs (ISO 14025, verify)	Advisory (not counted)
I35	Export control / regulated alloys	Regulatory	Ti alloys, Ni superalloys, Be, composites	Some alloys/forms are export-controlled or require end-use declarations	Jurisdiction-specific	Check regulations before export/import	—	Advisory (not counted)

Sheet 10 — Material Selection Report

Selection is multi-criteria: strength alone never decides. Cost entries are qualitative and unverified. Candidates are options, not recommendations without application data.

Application	Governing criteria	Candidate (condition)	Advantages	Trade-offs / risks	Required verification tests	Cost note (qualitative, unverified)
Robot chassis	Stiffness-to-weight, weldability/joining, cost, vibration	6061-T6 plate/extrusion	Good E/ρ, machinable, corrosion-resistant, widely available	Weld HAZ softening; lower fatigue strength than steel	Tensile/hardness on lot; weld NDT; modal/vibration test	Low–medium (qualitative, unverified)
Robot chassis	(as above)	5052-H32 / 5083 sheet	Formable, weldable, good corrosion resistance	Lower strength than 6061-T6	Bend test; weld inspection	Low–medium
Robot chassis	(as above)	CFRP laminate/sandwich	Highest specific stiffness; low mass	Cost, joining, damage tolerance, galvanic with Al	D3039/D7264/D2344 coupons; UT/thermography	High
Robot chassis	(as above)	IS 2062 E250 / A36 steel	Lowest cost, easy welding, stiff	Heavy; corrosion protection needed	Tensile/MTC; weld NDT	Low
Robotic arm	Stiffness (deflection), mass, fatigue, repeatability	7075-T6 / 6061-T6	Light, strong; 6061 weldable	7075 SCC-sensitive, not fusion-weldable	Tensile/hardness; conductivity temper check	Medium
Robotic arm	(as above)	CFRP tubes	High specific stiffness, damping	Impact damage, joints, cost	D3039/D7264; ILSS; NDT	High
Robotic arm	(as above)	Ti-6Al-4V	High specific strength, corrosion resistance	Low stiffness vs steel at same section, cost	Tensile, microstructure, fatigue	High
Shaft	Fatigue strength, torsion, notch sensitivity, wear/corrosion	AISI 1045 / C45 (Q&T)	Economical, machinable	Moderate hardenability	Tensile, hardness, impact if specified	Low
Shaft	(as above)	AISI 4140 / 42CrMo4 or 4340 (Q&T)	High strength/toughness; deeper hardening	Temper embrittlement/H2 risk at high strength	Tensile, Charpy, hardness; UT/MPI for critical bars	Medium
Shaft	(as above)	17-4 PH H1025 / 316L	Corrosion resistance (17-4 strong, 316L weak)	316L low strength; 17-4 SCC risk in some tempers	Tensile, hardness, corrosion test per service	Medium–high
Gear	Contact & bending fatigue, hardness, wear, quiet running	8620 / 20MnCr5 / 16MnCr5 (carburised)	Hard case + tough core	Distortion; case-depth control	Case depth, core hardness, microstructure, MPI after grinding	Medium
Gear	(as above)	18CrNiMo7-6 / 9310 (carburised)	Heavy-duty gears, high core toughness	Cost, heat-treat complexity	As above + cleanliness	Medium–high
Gear	(as above)	POM / PA66 (light duty)	Low noise, self-lubricating	Creep, temperature, moisture	Wear/creep test; D638 (conditioned)	Low
Bearing	Rolling-contact fatigue, hardness, cleanliness, dimensional stability	52100 / 100Cr6	Industry standard bearing steel	Limited corrosion resistance	Hardness, microstructure, cleanliness; bearing life test	Low–medium
Bearing	(as above)	440C stainless	Hardness + moderate corrosion resistance	Lower life than 52100 in many cases	Hardness, microstructure	Medium
Bearing	(as above)	Si3N4 hybrid / C93200 bushings / PTFE-PEEK liners	Si3N4: low density, non-magnetic; bronze: sliding; polymers: dry running	Ceramic brittleness; bronze load/speed limits	Weibull/proof test; wear test (G99 etc.)	Medium–high
Welded structure	Weldability, toughness, CE, thickness effects	IS 2062 E250/E350, A36, A572 Gr 50	Economical, code-based	Need quality designation, thickness and impact specs	MTC, CE, Charpy, weld NDT (MT/UT/RT)	Low
Welded structure	(as above)	304L/316L, 2205	Corrosion resistance	Sensitisation, HAZ issues, distortion	PMI, IGC/pitting tests where required	Medium–high
Welded structure	(as above)	5083 / 5052; 6061 (with HAZ reduction)	Light, good corrosion resistance	HAZ strength loss in 6061	Weld procedure qualification, hardness traverse	Medium
Corrosion-exposed component	Environment (Cl^-, pH, T), galvanic, SCC	316L / 2205 / 2507	PREN ranking; duplex stronger	SCC/pitting limits; cost	ASTM G48, A262 as applicable	Medium–high
Corrosion-exposed component	(as above)	Ti Grade 2 / Ni alloys (625, C-276, Monel 400) / C95400	Excellent seawater/chemical resistance	High cost; galvanic couples	Coupon immersion; crevice tests	High
High-temperature component	Creep, oxidation, thermal fatigue, cost	IN718 / IN625 / 310S / 321	IN718 high strength to ≈650 °C; 310S oxidation resistance	Cost; machinability; creep data required	Creep (E139), tensile at T, oxidation	High
High-temperature component	(as above)	H13 (hot-work tooling), SiC/Si3N4	H13 thermal-fatigue resistance; ceramics for extreme temperature	Brittleness of ceramics; H13 heat-check	Hardness, thermal-fatigue/shock tests	Medium–high
High-temperature component	(as above)	Mo / W (inert atmosphere only)	Very high melting point	Oxidise in air; brittle near RT	Atmosphere control, bend/tensile	High

Performance indices (formula-driven from Sheet 3) — stiffness- or strength-limited lightweight design

$E^{1/2}/\rho$: stiffness-limited light beam; $\sigma^{2/3}/\rho$: strength-limited light beam; E/ρ: stiffness-limited tie/column. CFRP is valid along fibres only. Strength = YS where available, else UTS. Higher is better.

Material	Condition	ρ (Mg/m³)	E (GPa)	Strength σ (MPa)	E/ρ	$E^{1/2}/\rho$	$\sigma^{2/3}/\rho$	σ/ρ (MPa per Mg/m³)	Rank $E^{1/2}/\rho$	Rank $\sigma^{2/3}/\rho$
ASTM A36	Plate/shapes (specified)	7.85	200.0	250	25.5	1.802	5.06	31.8	9	11
AISI 4140	Q&T (oil quench, ≈650 °C temper)	7.85	205.0	655	26.1	1.824	9.61	83.4	7	9
17-4 PH (UNS S17400)	Condition H1025 (A564 min)	7.78	197.0	1000	25.3	1.804	12.85	128.5	8	7
AA 6061-T6	Sheet/plate/extrusion (typical)	2.70	68.9	276	25.5	3.074	15.70	102.2	4	6
AA 7075-T6	Sheet/plate (typical)	2.81	71.7	503	25.5	3.013	22.51	179.0	5	2
AA 5083-H111	EN 485-2 minima	2.66	71.0	125	26.7	3.168	9.40	47.0	3	10
AZ31B	O-temper sheet (typical)	1.77	45.0	150	25.4	3.790	15.95	84.7	2	5
Ti-6Al-4V Gr 5	Annealed (B265 min)	4.43	113.8	825	25.7	2.408	19.86	186.2	6	3
CFRP UD epoxy (Vf≈0.6)	0° direction	1.56	135.0	1800	86.5	7.448	94.85	1153.8	1	1
UNS N07718	Solution + aged (AMS-type minimum, approx.)	8.19	200.0	1034	24.4	1.727	12.49	126.3	10	8
PEEK	Unfilled	1.30	3.6	100	2.8	1.460	16.57	76.9	11	4

Sheet 11 — Sources, Assumptions, Unresolved Items

Source	Used for	Official URL (home page — search by designation)	Access note
ASTM International	Standards catalogue (titles, editions, status)	https://www.astm.org/	Standard texts are paywalled; only catalogue entries accessible online — not accessed in this session
Bureau of Indian Standards (BIS)	IS catalogue / standards portal	https://www.bis.gov.in/	Search IS number in the BIS catalogue — not accessed in this session
ISO	ISO standards catalogue	https://www.iso.org/	Not accessed in this session
ASM International	ASM Handbooks (property data)	https://www.asminternational.org/	Values recalled from general handbook knowledge, not re-fetched
MatWeb	Typical material datasheets	https://www.matweb.com/	Values recalled, not re-fetched
NIST	Reference data	https://www.nist.gov/	Not accessed
The Aluminum Association	Alloy/temper designations	https://www.aluminum.org/	Not accessed
SSAB (Hardox)	Manufacturer datasheets	https://www.ssab.com/	Manufacturer values change by revision; verify current datasheet
Special Metals	Ni-alloy datasheets	https://www.specialmetals.com/	Not accessed

Assumptions & status definitions

A1	No network or authoritative-document access was available in this session; all property values and standard information were recalled from training knowledge and marked Conditionally verified or Unverified accordingly. Nothing is marked 'Verified' except arithmetic and unit checks.
A2	'Verified' = checked against an authoritative source or independently recalculated; 'Conditionally verified' = consistent with recalled handbook/standard information but not checked against the document; 'Unverified' = cannot be confirmed or under-specified; 'Incorrect' = supplied claim contains an error.
A3	Values labelled Typical are handbook-type figures; values labelled Specified minimum come from product standards and are not guaranteed design allowables.
A4	All property values are room-temperature unless stated; test temperature column supplied.
A5	Specific strength uses UTS if present, otherwise YS; specific stiffness uses E. Shear modulus is derived assuming isotropy.
A6	Hall–Petch, Goodman, Archard and Larson–Miller worked examples use illustrative assumed parameters (labelled) — not fitted to a specific dataset.
A7	Gap-list grades are suggestions and are excluded from coverage counts.
A8	Cost entries are qualitative and unverified.

Unresolved items

U1	Confirm current/withdrawn status of ASTM E164.
U2	Confirm current edition/revision status of IS 1865 and all IS edition years in Standards sheet.
U3	Check exact current titles (E165/E165M, E1444/E1444M, E1742/E1742M) in the ASTM catalogue.
U4	Re-fetch grade-specific property values from ASM/manufacturer datasheets and record publication dates.

U5	Resolve under-specified materials (ceramics, composites, CoCr, graphite, CuBe) by choosing exact grades.
U6	Add fatigue, impact and fracture-toughness values with test conditions for critical grades.

Colour legend

Sourced / recalled value

Assumed input (editable)

Linked from another sheet

Calculated by formula